

Modelling Financial Markets

Chapter 7: Value-at-Risk and Expected Shortfall



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How Large Are Tomorrow's Tail Losses, and How Precisely Can We Tell?

What we will explain

- Two tail measures: VaR (Value-at-Risk), a loss threshold, and ES (Expected Shortfall), the mean loss beyond it
- Part I: definitions, coherence, unconditional estimation methods and their precision
- Part II: conditional forecasts (filtered historical simulation, FHS, and direct quantile models, CAViaR), extreme value theory, horizon, portfolio, regulation, a case study and one pass of the research loop assisted by AI (artificial intelligence)
- Solved exercises, Seminar 7: A1, A2, A3, A5, B1

What you should be able to do

- define VaR and ES and explain why ES is coherent and VaR is not
- estimate both historically, parametrically, by filtered simulation and by extreme value theory, each with a standard error or an interval
- scale them to 10 days and split them across assets
- read the Basel rules, including FRTB (Fundamental Review of the Trading Book, the 2019 market-risk standard)

Which Losses and Units Will We Use?

- Running example: S&P 500 daily losses, 2000–2026; comparisons on BET (main index of the Bucharest Stock Exchange, BVB), Bitcoin, EUR/RON (reference rate of the National Bank of Romania, BNR) and gold; data up to 18 September 2026 (full inventory in the Appendix)
- Units: daily log returns in %, $r_t = 100\,(\ln P_t - \ln P_{t-1})$, with P_t the closing price on day t

How Much Can We Lose? RiskMetrics and VaR



J.P. Morgan & Co. building, 23 Wall Street, New York

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- Tomorrow's portfolio value is random; management and regulators need **one number**: a loss exceeded only rarely, to set limits and **capital**
- J.P. Morgan published **RiskMetrics** in October 1994, a free methodology and data set for market risk; its Technical Document (4th ed., 1996) became the reference for VaR J.P. Morgan/Reuters (1996)
- Textbooks: Franke, Härdle & Hafner (2019), McNeil, Frey & Embrechts (2015), Daníelsson (2011)

Losses, Horizons and Tail Probabilities

- Return over one day: $X_{t+1} = r_{t+1}$, the log return in %; **loss** $L_{t+1} = -X_{t+1}$, a **log loss**
  - Positive L = the position lost money; negative L = a gain
  - In money, with V_t the value of the position: loss $= V_t\,(1 - e^{-L_{t+1}/100}) \approx V_t L_{t+1}/100$ for small L; with simple returns R, $V_t\,(-R_{t+1})/100$ is exact
  - Example: a price falling from 100 to 90 is a simple return of -10% but a log return of $100\ln(0.9) = -10.54\%$; log returns add over days, simple returns add across assets
  - VaR converts exactly through this monotone map; ES must be converted scenario by scenario, then averaged
- **Horizon** h: the number of days over which the loss is measured (1 day, 10 days)
- **Level** α: the **tail probability**, typically $\alpha = 1\%$ or 2.5%: "VaR 1%", "ES 2.5%"
- **Sign convention in this course**: VaR and ES are **positive** numbers when they are losses

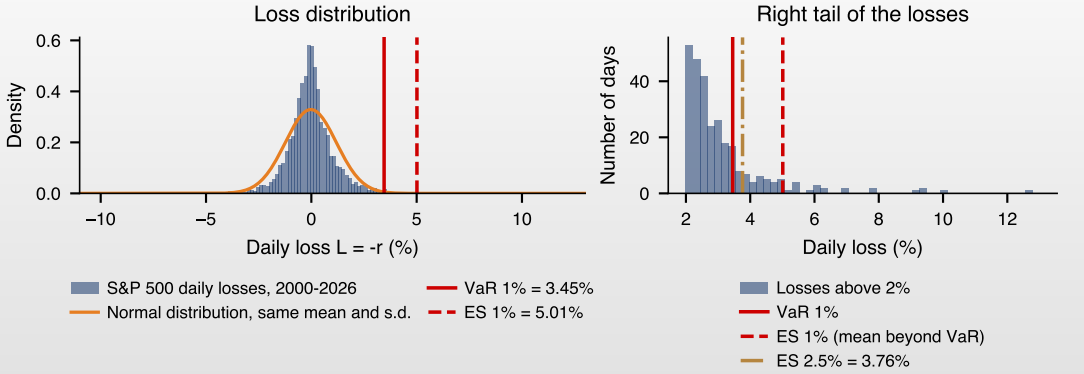
Definitions (1/2): Value-at-Risk

- **VaR** at level α: the loss exceeded with probability α (exactly for continuous L, at most α in general)
  - $\mathrm{VaR}_\alpha(X) = -\inf\{x \in \mathbb{R} : P(X \leq x) > \alpha\} = -q_\alpha(X)$
  - In words: VaR is minus the α-quantile of the return, i.e. the return threshold below which only a fraction α of days fall, with the sign changed
- Notation
  - X: the return of the position over the horizon; $P(X \leq x)$: the probability that it is at most x, i.e. the distribution function at x
  - $\inf\{\cdot\}$: the infimum (largest lower bound) of the set, here the smallest x at which the distribution function exceeds α
  - $q_\alpha(X)$: the α-quantile of X, negative for small α; for a continuous loss $L = -X$: $\mathrm{VaR}_\alpha = q_{1-\alpha}(L)$, the upper quantile of the loss
- Reading of VaR 1%
  - "With probability 1%, tomorrow's loss will exceed $\mathrm{VaR}_{1\%}$"
  - VaR does **not** say how large the loss is beyond this threshold

Definitions (2/2): Expected Shortfall

- **Expected Shortfall**: the average of all quantiles in the tail of probability α
  - $\mathrm{ES}_\alpha(X) = -\frac{1}{\alpha}\int_0^\alpha q_u(X)\,du = \frac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_u(X)\,du$
  - u: the integration variable, a tail probability running from 0 to α; the integral divided by α is the average of VaR_u over all levels $u \le \alpha$
- For a continuous distribution: the mean loss on the days beyond VaR
  - $\mathrm{ES}_\alpha(X) = -\mathbb{E}[X \mid X \le q_\alpha(X)]$; $\mathbb{E}[\cdot \mid \cdot]$: the conditional expectation, here the mean of X over the worst α fraction of outcomes
  - Always $\mathrm{ES}_\alpha \ge \mathrm{VaR}_\alpha$: ES looks **into** the tail, VaR only at its entrance; two portfolios with the same VaR can have very different ES
- Other names: Conditional VaR (CVaR), Tail VaR, Average VaR Rockafellar & Uryasev (2002), Acerbi & Tasche (2002)

VaR and ES on the S&P 500 (1/2)



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VaR and ES on the S&P 500 (2/2)

- Daily losses 2000–2026 against the Normal distribution with the same mean and standard deviation (left); the tail (right)
- Historical $\mathrm{VaR}_{1\%} = 3.45\%$: 68 of 6 717 days (1.01%) lost more; on those days the average loss, $\mathrm{ES}_{1\%}$, was 5.01%
- At the 2.5% level: historical VaR is 2.51% and ES is 3.76%; largest loss 12.77% on 2020-03-16 (COVID-19 crash), about ten standard deviations
- The Normal distribution fits the centre but is far too thin in the tail (Chapter 1)

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The Normal Distribution: ES in One Line (1/2)

- Setting: $X \sim N(\mu, \sigma^2)$, a Normal return with mean μ and standard deviation σ
  - Φ, φ: the distribution function and the density of the standard Normal distribution $N(0,1)$
  - $z_\alpha = \Phi^{-1}(\alpha) < 0$: the standard Normal α-quantile, e.g. $z_{1\%} = -2.326$
- VaR: the mean minus $|z_\alpha|$ standard deviations, sign changed
  - $\mathrm{VaR}_\alpha = -(\mu + \sigma z_\alpha)$
- ES: the same form, with the larger factor $\varphi(z_\alpha)/\alpha$ in place of $-z_\alpha$
  - $\mathrm{ES}_\alpha = -\mu - \dfrac{\sigma}{\alpha} \displaystyle\int_{-\infty}^{z_\alpha} u\varphi(u)\, du = -\mu + \sigma \dfrac{\varphi(z_\alpha)}{\alpha}$
  - The integral divided by α is the mean of the standardised shock u over the tail $u \le z_\alpha$; the integral itself equals $-\varphi(z_\alpha)$ because $\varphi'(u) = -u\varphi(u)$

The Normal Distribution: ES in One Line (2/2)

- Constants to remember
  - $-z_{1\%} = 2.326$, $\varphi(z_{1\%})/0.01 = 2.665$
  - $-z_{2.5\%} = 1.960$, $\varphi(z_{2.5\%})/0.025 = 2.338$
  - So, under the Normal distribution, ES 2.5% is almost equal to VaR 1% (2.338 against 2.326 standard deviations)
- S&P 500 ($\hat{\mu} = 0.025\%$, $\hat{\sigma} = 1.213\%$; the hat marks a value estimated from the data): Normal $\mathrm{VaR}_{1\%} = 2.80\%$ vs historical 3.45%
  - Arithmetic: $\mathrm{VaR}_{1\%} = -\hat{\mu} + \hat{\sigma}\,(-z_{1\%}) = -0.025 + 1.213 \times 2.326 \approx 2.80\%$
  - The historical VaR is 1.24 times the Normal VaR
- Under the Normal distribution, for a given μ, VaR and ES carry no information beyond σ

The Student-t Distribution: VaR and ES

- Model: the return is a shifted and rescaled Student-t variable, $X = m + s\,T$ with $T \sim t_\nu$
  - m: location; $s > 0$: scale; ν: degrees of freedom, smaller ν = heavier tails
  - $t_\alpha = t_\nu^{-1}(\alpha) < 0$: the α-quantile of t_ν; g_ν: the density of t_ν
- VaR as for the Normal distribution, with t_α in place of z_α; ES gets an extra factor $(\nu + t_\alpha^2)/(\nu - 1) > 1$ for the heavy tail
  - $\mathrm{VaR}_\alpha = -(m + s\,t_\alpha), \qquad \mathrm{ES}_\alpha = -m + s\,\dfrac{g_\nu(t_\alpha)}{\alpha}\,\dfrac{\nu + t_\alpha^2}{\nu - 1}$ (for $\nu > 1$)
  - Variance $s^2\nu/(\nu - 2)$ exists only for $\nu > 2$, so s is not the standard deviation

Student-t Fit to the S&P 500, Step by Step

- Maximum likelihood estimates on daily returns in %: $\hat{m} = 0.068$, $\hat{s} = 0.702$, $\hat{\nu} = 2.666$
  - Implied standard deviation $\hat{s}\sqrt{\hat{\nu}/(\hat{\nu}-2)} = 1.40\%$, not the sample $\hat{\sigma} = 1.21\%$: the scale $\hat{s}$, not $\hat{\sigma}$, enters the formulas
- **VaR**: quantile $t_{1\%} = t_{\hat{\nu}}^{-1}(0.01) = -5.028$
  - $\mathrm{VaR}_{1\%} = -(\hat{m} + \hat{s}\, t_{1\%}) = -0.068 + 0.702 \times 5.028 \approx 3.46\%$
- **ES**: density $g_{\hat{\nu}}(t_{1\%}) = 0.00490$, so $g_{\hat{\nu}}(t_{1\%})/\alpha = 0.490$; tail factor $(\hat{\nu} + t_{1\%}^2)/(\hat{\nu} - 1) = 16.77$
  - $\mathrm{ES}_{1\%} = -\hat{m} + \hat{s}\,\dfrac{g_{\hat{\nu}}(t_{1\%})}{\alpha}\,\dfrac{\hat{\nu} + t_{1\%}^2}{\hat{\nu} - 1} = -0.068 + 0.702 \times 0.490 \times 16.77 \approx 5.71\%$
- Why ES overshoots: with $\hat{\nu}$ close to 2 the tail factor is large, so ES is 5.71% against the historical 5.01%, while VaR matches

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How Do We Compare Tail Shapes Fairly?

- To isolate tail shape, hold the mean and the standard deviation fixed
- Normal and Student-t models fitted to the same data need not have the same scale parameter
- Example: zero mean, standard deviation 2% and $\nu = 4$; the Student-t scale is $s = 2\sqrt{(4-2)/4} = 1.414\%$, the Normal scale is 2%
- With $s = 2\%$ in both models the Student-t standard deviation would be larger: the difference would mix a scale effect with a tail-shape effect
- The controlled comparison (same mean, same standard deviation) shows the effect of tail shape (Seminar 7, A2)
- Fitting the two models separately to the same data also changes the scale: the differences in the Estimation Methods section mix both effects

Coherent Risk Measures (1/2): the Four Axioms

- A **risk measure** ρ maps a loss L (a random variable) to a number, the capital to hold; it is **coherent** if it satisfies four axioms Artzner, Delbaen, Eber & Heath (1999)
  - L_1, L_2: two losses; c: a sure amount (constant); $\lambda \geq 0$: a scaling factor (position size)
- **Monotone**: $L_1 \leq L_2 \Rightarrow \rho(L_1) \leq \rho(L_2)$
  - A position that always loses more needs more capital
- **Translation invariant**: $\rho(L + c) = \rho(L) + c$
  - Adding a sure loss c raises the capital by exactly c
- **Positively homogeneous**: $\rho(\lambda L) = \lambda \rho(L)$
  - Doubling the position doubles the capital
- **Subadditive**: $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$
  - Merging two positions never needs more capital than holding them apart: diversification does not add risk

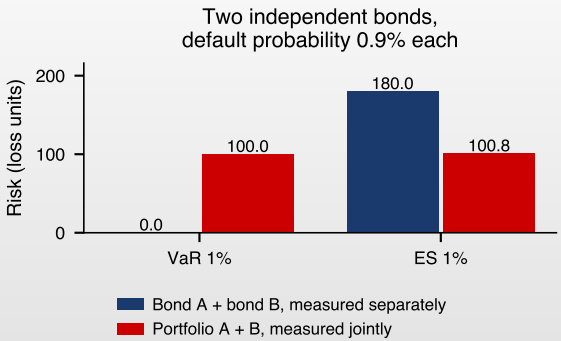
Coherent Risk Measures (2/2): Why the Axioms Matter

- **Limits**
  - By subadditivity, bank-wide risk does not exceed the sum of the desk risk limits, $\sum_i \rho_i$
  - **No regulatory arbitrage** by splitting a firm
- **Optimisation**: homogeneity + subadditivity $\Rightarrow$ convexity in the portfolio weights
  - Every local minimum is global; uniqueness would need **strict** convexity, which scenario-based ES lacks (it is piecewise linear)
- VaR satisfies the first three axioms, but **not** subadditivity in general
  - Exception: jointly elliptical losses (multivariate Normal or Student-t) and $\alpha < 50\%$, where VaR and ES are both $\mu_p + \sigma_p \times$ constant McNeil, Frey & Embrechts (2015)
  - μ_p, σ_p: mean and standard deviation of the portfolio loss (heavy-tailed refinements: Appendix)

Counterexample: Two Bonds, Step by Step

- Bonds A and B, independent; each loses 100 with probability $p = 0.9\%$ (default) and 0 otherwise; X: the value change, $X = -L$
- General formula for a discrete loss $L = -X$: $\mathrm{ES}_\alpha = \frac{1}{\alpha}\left(\mathbb{E}[L\,1\{L > \mathrm{VaR}_\alpha\}] + \mathrm{VaR}_\alpha\left(\alpha - P(L > \mathrm{VaR}_\alpha)\right)\right)$
  - $1\{\cdot\}$: indicator, 1 if the condition holds and 0 otherwise; first term: losses strictly beyond VaR; second term: the rest of the tail probability α, placed at VaR itself
- **One bond**: $P(X \le -100) = p = 0.9\%$, not above 1%, so $q_{1\%} = 0$ and $\mathrm{VaR}_{1\%}(A) = \mathrm{VaR}_{1\%}(B) = 0$
  - $\mathrm{ES}_{1\%}(A) = (100 \cdot 0.009 + 0 \cdot 0.001)/0.01 = 90$
- **Portfolio A + B**: $P(X \le -100) = 1 - (1-p)^2 = 1.79\% > 1\%$, $P(X \le -200) = p^2 = 0.0081\%$, so $q_{1\%} = -100$
  - $\mathrm{VaR}_{1\%}(A+B) = 100 > 0 + 0 = \mathrm{VaR}_{1\%}(A) + \mathrm{VaR}_{1\%}(B)$
- $\mathrm{ES}_{1\%}(A+B) = 100.81 \le 180 = \mathrm{ES}_{1\%}(A) + \mathrm{ES}_{1\%}(B)$

Subadditivity: VaR Fails, ES Holds



- Blue: sum of the two stand-alone measures; red: the measure of the combined portfolio
- Each bond alone "hides" its default in the last 1%; together, at least one default has probability $1.79\% > 1\%$ and VaR jumps to 100
- A VaR limit would reward **concentration**: holding only A looks riskless; ES sees the size of the tail and is coherent Acerbi & Tasche (2002)
- Credit portfolios, options and other skewed pay-offs are where VaR fails in practice

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Quick Check: Which Position Is Riskier?

- Two positions, losses in %, level $\alpha = 10\%$
- **A**: loss L_A uniform on $[0, 10]$, every value equally likely
  - Find $\mathrm{VaR}_{10\%}(A)$ and $\mathrm{ES}_{10\%}(A)$ from the definitions
- **B**: loss 0 with probability 0.95, loss 20 with probability 0.05
  - Find $\mathrm{VaR}_{10\%}(B)$ and $\mathrm{ES}_{10\%}(B)$ with the discrete ES formula of the two-bond slide
- Which position does each measure call riskier, and what does VaR miss? Answers in the Appendix

ES as an Optimisation Problem: Rockafellar–Uryasev

- **Theorem** Rockafellar & Uryasev (2000): for any integrable loss L
  - $\mathrm{ES}_\alpha(L) = \min_{v \in \mathbb{R}} F(v), \quad F(v) = v + \frac{1}{\alpha}\mathbb{E}[(L - v)_+], \quad (x)_+ = \max(x, 0)$
  - v: a candidate threshold; $F(v)$: the threshold plus the expected excess of the loss over it, scaled up by $1/\alpha$ (Seminar 7 writes the same function as $G_\alpha(v)$)
  - The minimiser is $\mathrm{VaR}_\alpha(L)$, the minimum value is $\mathrm{ES}_\alpha(L)$; $F(\mathrm{VaR}_\alpha)$ is the discrete ES formula of the two-bond slide
- F is convex in v and, for a portfolio loss $-w^\top R$, jointly convex in (w, v)
  - w: vector of weights; R: vector of asset returns
  - ES is convex in the weights, which **proves** subadditivity (proof in the Appendix)
  - With historical scenarios, minimising ES becomes a linear program (the Portfolio VaR section)
- S&P 500 losses, $\alpha = 2.5\%$: $\arg\min F = 2.51\%$ (the minimising v), $\min F = 3.76\%$
  - Historical VaR and ES (interpolated quantile, tail average): 2.51% and 3.76%
  - Equal only after rounding: the empirical minimiser is an order statistic

Coherence versus Robustness

- **Qualitative robustness**: a small change in the data distribution changes the law of the estimator only a little Cont, Deguest & Scandolo (2010)
  - Historical VaR is qualitatively robust at distributions with a unique α-quantile; historical ES is **not**: one extreme loss moves it without bound (the precision section)
- S&P 500, last 500 days: drop the single largest loss (6.16%, 2025-04-04)
  - $\mathrm{ES}_{2.5\%}$: 2.94% $\to$ 2.60%; $\mathrm{VaR}_{1\%}$: 2.73% $\to$ 2.68%
- A trade-off, not a free lunch: subadditivity needs the tail beyond the quantile, which is exactly what makes ES sensitive to single observations
- FRTB chose coherence for capital (ES 2.5%) and kept robust, elicitable VaR for backtesting

Elicitability: Why ES Is Harder to Evaluate (Chapter 8)

- A risk measure is **elicitable** if it is the minimiser of an expected scoring function
  - VaR: the quantile $q_\alpha(X)$ minimises the pinball loss $\mathbb{E}[(1\{X \le v\} - \alpha)(v - X)]$
  - Scoring function: a penalty computed from the forecast v and the outcome X, lower is better; the pinball loss charges $1 - \alpha$ per unit when X falls below v and α per unit when it stays above
- ES alone is **not** elicitable Gneiting (2011): no strictly consistent score ranks ES forecasts on their own
- The pair (VaR, ES) is jointly elicitable Fissler & Ziegel (2016): joint scores such as FZ0 (the zero-homogeneous Fissler–Ziegel score of Patton, Ziegel & Chen (2019)) rank (VaR, ES) forecasts (▸ Appendix; Chapter 8)
- Consequence: comparing ES forecasts is harder than comparing VaR forecasts

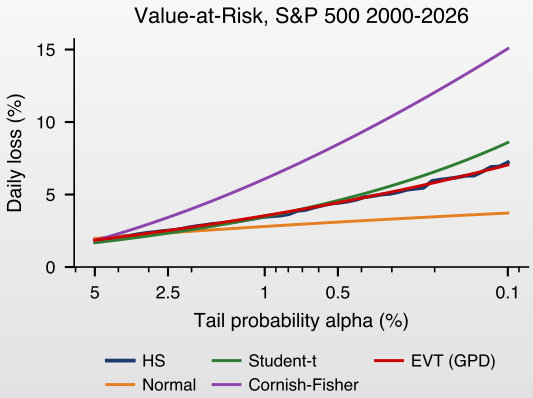
Historical Simulation

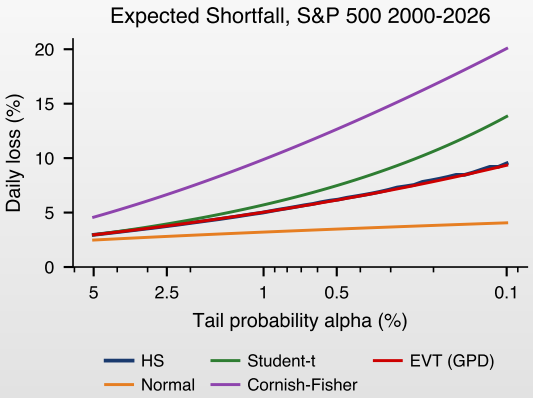
- Take the last n daily losses $L_{t-n+1}, \ldots, L_t$ and sort them: $L_{(1)} \leq \cdots \leq L_{(n)}$; $L_{(i)}$: the i-th smallest loss (the i-th order statistic)
  - $\widehat{\mathrm{VaR}}_\alpha$ = empirical $(1-\alpha)$-quantile, interpolated linearly at position $1 + (n-1)(1-\alpha)$ of the sorted losses (the convention used throughout)
  - $\widehat{\mathrm{ES}}_\alpha$ = mean of the k_α losses $\geq \widehat{\mathrm{VaR}}_\alpha$, $k_\alpha \approx n\alpha$; the integral definition gives the last included loss the weight $n\alpha - \lfloor n\alpha \rfloor$ ($\lfloor x \rfloor$: the integer part of x), and the simple mean of the k_α losses equals it only when $n\alpha$ is an integer
- Choice of window n: 250 days reacts faster, 1000 days is more precise
  - With $n = 500$: 5 losses lie beyond VaR 1%; ES 2.5% averages $k = 13$ losses ($n\alpha = 12.5$)
- No loss beyond the worst one in the window can appear in the estimate
- **Ghost effect**: a crash day enters the window, stays for n days, then drops out and VaR falls abruptly

Cornish–Fisher: Correcting the Normal Quantile, and Where It Breaks

- Adjust the Normal quantile for **skewness** $S = \mathbb{E}[(X-\mu)^3]/\sigma^3$ and **excess kurtosis** $K = \mathbb{E}[(X-\mu)^4]/\sigma^4 - 3$ Cornish & Fisher (1938)
  - $\tilde{z}_\alpha = z_\alpha + \frac{(z_\alpha^2-1)S}{6} + \frac{(z_\alpha^3-3z_\alpha)K}{24} - \frac{(2z_\alpha^3-5z_\alpha)S^2}{36}, \quad \mathrm{VaR}_\alpha = -(\mu+\sigma\tilde{z}_\alpha)$
  - $\tilde{z}_\alpha$: the corrected quantile, the Normal quantile $z_\alpha < 0$ plus one term for skewness, one for excess kurtosis and one for squared skewness; $S = K = 0$ gives back the Normal VaR
  - For α below about 4% ($|z_\alpha| > \sqrt{3}$): $S < 0$ (long left tail of returns) or $K > 0$ (heavy tails) push $\tilde{z}_\alpha$ further below z_α and raise VaR; the S^2 term pushes the other way. At $\alpha = 5\%$ the K term has the opposite sign
- ES: average of $-(\mu+\sigma\tilde{z}_u)$ over $u \in (0,\alpha)$, numerically
- An expansion around the Normal distribution: accurate only for **small** S and K; for large K the map $z \mapsto \tilde{z}$ is not even monotone (Seminar 7, A6)
- S&P 500 returns: $S = -0.35$, $K = 10.7$: far outside that range
  - $\mathrm{VaR}_{1\%}^{CF} = 6.08\%$ versus historical 3.45%: the kurtosis term overshoots

Five Methods Across Tail Probabilities (1/2)





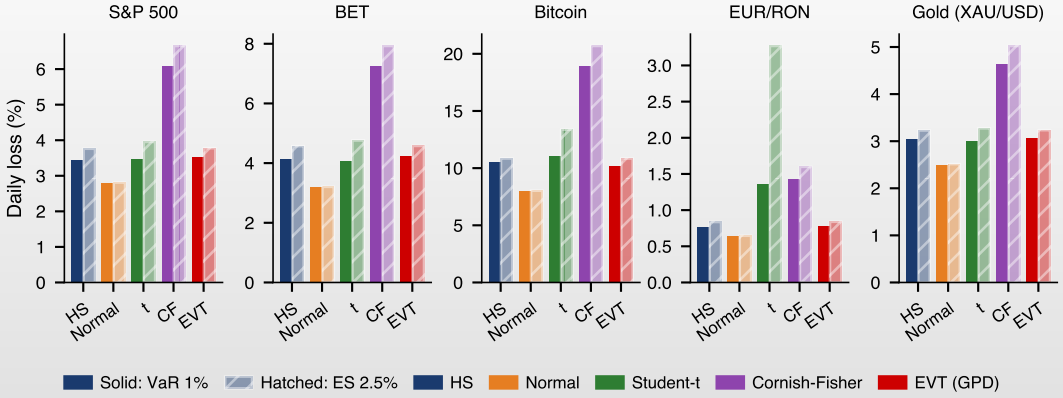
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Five Methods Across Tail Probabilities (2/2)

- S&P 500 daily losses 2000–2026, α from 5% down to 0.1%; legend: HS = historical simulation; EVT (extreme value theory) = a GPD (generalised Pareto distribution) tail, a preview of the Extreme Value Theory section
- VaR 0.1% (one day in four years): historical 7.22%, EVT 7.04%, Student-t 8.58%, Cornish–Fisher 15.06%, Normal 3.72%, about half
- The gap between methods **grows** as α falls, more for ES than for VaR; EVT and historical agree down to 0.5%, below that EVT extrapolates a fitted tail

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Methods Compared on Five Assets (1/2)



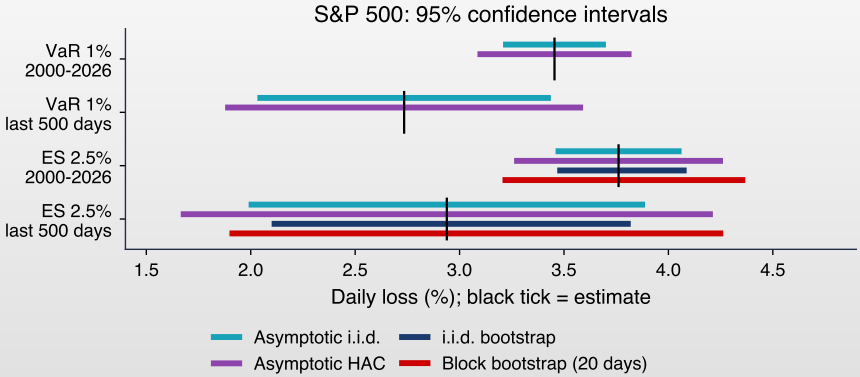
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Methods Compared on Five Assets (2/2)

- Solid bars: $\mathrm{VaR}_{1\%}$; hatched bars: $\mathrm{ES}_{2.5\%}$; daily losses in %, full samples (all numbers in the Appendix); bar labels: CF = Cornish–Fisher, t = Student-t; XAU/USD: the gold price in US dollars per troy ounce
- Normal: too low everywhere (Bitcoin $\mathrm{VaR}_{1\%}$ 7.99% vs 10.51% historical)
- Cornish–Fisher: overshoots, since excess kurtosis exceeds 7 in every series (BET 7.25% vs 4.13%)
- Student-t fails on EUR/RON: $\hat{\nu} = 1.25$ (infinite variance) gives an unrealistic ES 2.5% of 3.27%, against 0.84% historical
  - Many near-zero reference-rate changes drive $\hat{\nu}$ down
- EVT stays close to historical simulation for VaR 1% in all five series

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Precision on the S&P 500 (1/2)



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Precision on the S&P 500 (2/2)

- 95% intervals (bars), with estimates marked by black ticks; bootstrap: 2 000 resamples (Seminar 7, B2); complete table in the ▸ Appendix
- Full sample: clustering widens the ES interval from 0.60 (i.i.d.) to 1.00 (HAC), as the bootstrap does (0.62 i.i.d. vs 1.16 by blocks); last 500 days: ES 2.5% averages 13 losses, HAC interval width 2.55 pp
- These intervals cover the **unconditional** VaR and ES; tomorrow's risk is a conditional question (next section)

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Two Usable Standard Errors (1/2): VaR

- Standard error (SE): the standard deviation of an estimator across repeated samples; a 95% interval is roughly the estimate ± 1.96 SE
- **VaR**: i.i.d. (independent and identically distributed) losses with density f continuous and positive at VaR_α
  - $\mathrm{SE}(\widehat{\mathrm{VaR}}_\alpha) \approx \dfrac{1}{f(\mathrm{VaR}_\alpha)} \sqrt{\dfrac{\alpha(1-\alpha)}{n}}$
  - $\sqrt{\alpha(1-\alpha)/n}$: the SE of the share of days beyond VaR; dividing by the density $f(\mathrm{VaR}_\alpha)$ turns it into an error on the loss scale
  - The tail density is small, so the SE is large
  - $\sigma = 1.2\%$, $n = 500$: SE of $\mathrm{VaR}_{1\%} = 0.20$ percentage points (pp) under the Normal distribution, 0.43 pp under a t_4 with the same σ
- t_4: the Student-t distribution with 4 degrees of freedom

Two Usable Standard Errors (2/2): ES

- **ES** Chen (2008): influence function $\mathrm{IF}(x) = \mathrm{VaR}_\alpha + (x - \mathrm{VaR}_\alpha)_+/\alpha - \mathrm{ES}_\alpha$, unbounded in x
  - $\mathrm{IF}(x)$: the effect of one extra observation x on the estimate; it grows without bound with x, so one extreme loss can move ES arbitrarily far
- The standard error
  - If $\mathbb{E}[L^2] < \infty$: $\mathrm{SE}(\widehat{\mathrm{ES}}_\alpha) \approx \frac{1}{\alpha}\sqrt{\frac{\mathrm{Var}((L - \mathrm{VaR}_\alpha)_+)}{n}}$; Var: the variance; estimating VaR adds no first-order term (Rockafellar–Uryasev)
  - For Pareto-type tails the **relative** error falls like $1/\sqrt{n\alpha}$: the tail count governs precision Pele & Mazurencu-Marinescu-Pele (2026)
- Volatility clustering: calm days and turbulent days come in runs
  - Replace the variances by long-run HAC (heteroskedasticity and autocorrelation consistent) variances Newey & West (1987)of $1\{L_t > \mathrm{VaR}_\alpha\}$ and $(L_t - \mathrm{VaR}_\alpha)_+$
  - Or use a block bootstrap (derivations: Appendix)

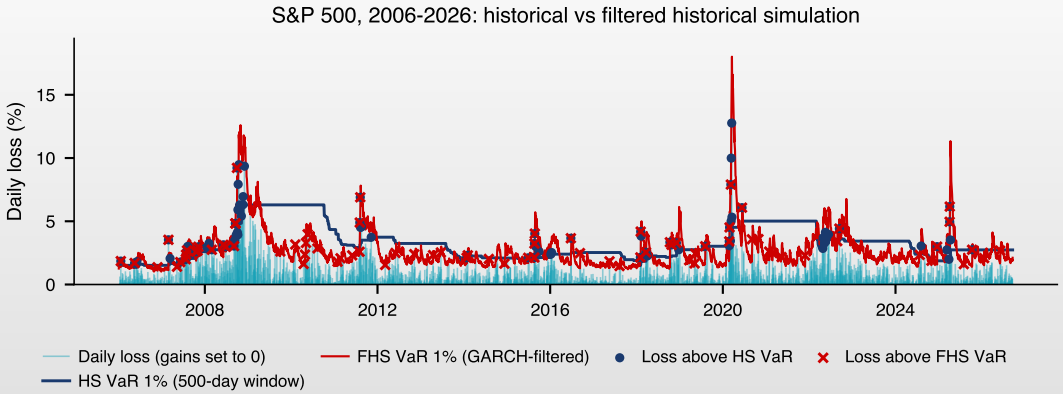
Bootstrap and HAC: How the Intervals Are Built

- **Bootstrap** (i.i.d.): draw n losses with replacement from the sample and recompute $\widehat{\mathrm{VaR}}$ or $\widehat{\mathrm{ES}}$
  - Repeat $B = 2\,000$ times; the 95% interval runs from the 2.5% to the 97.5% quantile of the B recomputed values
- **Moving-block bootstrap**: draw blocks of 20 consecutive days instead of single days and join $n/20$ of them
  - Calm and turbulent stretches stay intact inside each block, so every resample keeps the volatility clustering
- **HAC long-run variance** of a series y_t (here $1\{L_t > \mathrm{VaR}_\alpha\}$ or $(L_t - \mathrm{VaR}_\alpha)_+$): $\hat\gamma_0 + 2\sum_{j=1}^{m}\left(1 - \frac{j}{m+1}\right)\hat\gamma_j$
  - $\hat\gamma_j$: sample autocovariance of y_t at lag j, $\hat\gamma_0$ its variance; Bartlett weights $1 - j/(m+1)$ shrink distant lags; lag $m = \lfloor 4(n/100)^{2/9} \rfloor$ ($m = 10$ for the full S&P 500 sample)
  - It replaces the i.i.d. variance $\hat\gamma_0$ in the SE formulas. Toy check: $\hat\gamma_0 = 1$, $\hat\gamma_1 = 0.3$, all other $\hat\gamma_j = 0$, $m = 1$: long-run variance $1 + 2 \times 0.5 \times 0.3 = 1.3$, so the SE grows by $\sqrt{1.3} \approx 1.14$
- Worked numbers for the BET: Seminar 7, B1 (i.i.d., HAC and block-bootstrap intervals for VaR 1%)

Quick Check: How Many Days Make a Precise VaR?

- i.i.d. Normal daily losses with mean 0 and $\sigma = 1\%$; level $\alpha = 5\%$, so $\mathrm{VaR}_{5\%} = 1.645\%$ and the density there is $f(\mathrm{VaR}_{5\%}) = \varphi(1.645)/\sigma = 0.1031$
  - (a) With $n = 1\,000$ days, what is the SE of historical $\mathrm{VaR}_{5\%}$ and the half-width of its 95% interval?
  - (b) How many days halve that SE?
  - (c) With $n = 1\,000$, how many losses does historical $\mathrm{ES}_{1\%}$ average, and why is that a warning?
- Answers in the Appendix

Historical vs Filtered Historical Simulation: S&P 500 (1/2)



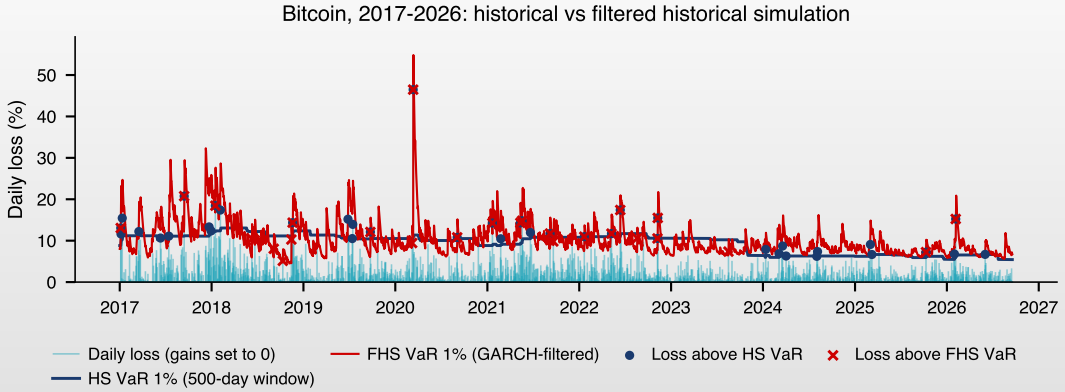
MFM_ch7_hs_fhs_sp500

Historical vs Filtered Historical Simulation: S&P 500 (2/2)

- One-day-ahead $\mathrm{VaR}_{1\%}$; GARCH re-estimated each January on all earlier data, so each forecast uses only information available at its origin
- Days with a loss above VaR 1%, 2006–2026 (5 210 days; target 1%): historical 500 days 1.67%, filtered 1.54%, GARCH with Normal shocks 2.34%
- Historical exceedances **cluster** (2008, 2020): the window reacts months late; filtered VaR peaked at 18.01% on 2020-03-18 and spreads exceedances over time (tests in Chapter 8)

MFM_ch7_hs_fhs_sp500

Historical vs Filtered Historical Simulation: Bitcoin (1/2)



MFM_ch7_hs_fhs_btc

Historical vs Filtered Historical Simulation: Bitcoin (2/2)

- Same design, forecasts from January 2017, 7 days a week; days with a loss above VaR (3 548 days): historical 1.16%, filtered 0.76%, GARCH with Normal shocks 1.58%
- Filtered VaR peaked at 54.83% in March 2020, after the crash of 2020-03-12: a log loss of 46.47%, i.e. a price fall of 37.17% (log and simple losses differ for large moves)
- Volatility has fallen since 2022 and the 500-day window follows with a long delay

MFM_ch7_hs_fhs_btc

Filtered Historical Simulation (1/2): the Model

- Idea Barone-Adesi, Giannopoulos & Vosper (1999), Hull & White (1998): history gives the **shape** of the shocks, a GARCH (generalised autoregressive conditional heteroskedasticity) model gives today's **scale**
  - $r_t = \mu_t + \sigma_t z_t$, $\mu_t = c + \phi r_{t-1}$, $\sigma_t^2 = \omega + \alpha_1 (r_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2$ (Chapter 5, Bollerslev (1986); α_1 is a GARCH coefficient, not the level α)
- Notation
  - μ_t: the conditional mean, an AR(1) (autoregressive of order 1) in the previous return with constant c and coefficient ϕ
  - σ_t: the conditional volatility; z_t: i.i.d. shock with mean 0 and variance 1, distribution left unspecified
  - $\omega > 0$: constant; $\alpha_1 \geq 0$: reaction to yesterday's squared surprise; $\beta_1 \geq 0$: weight on yesterday's variance; $\alpha_1 + \beta_1$: persistence; $\alpha_1 + \beta_1 < 1$ ensures a finite unconditional variance (close to 1: shocks fade slowly)

Filtered Historical Simulation (2/2): Three Steps

- Step 1: estimate the AR(1)-GARCH(1,1) model by quasi-maximum likelihood (Normal likelihood, robust to other shock distributions)
- Step 2: standardised residuals $\hat{z}_t = (r_t - \hat{\mu}_t)/\hat{\sigma}_t$: approximately i.i.d., still heavy-tailed
- Step 3: rescale by tomorrow's volatility
  - $\mathrm{VaR}_{\alpha,t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\,\hat{\xi}_\alpha$, $\hat{\xi}_\alpha = -q_\alpha(\hat{z})$; $\quad \mathrm{ES}_{\alpha,t+1} = -\hat{\mu}_{t+1} - \hat{\sigma}_{t+1}\,\bar{z}_\alpha$
  - $\hat{\mu}_{t+1}$, $\hat{\sigma}_{t+1}$: the model forecasts for tomorrow; $\hat{\xi}_\alpha$: minus the empirical α-quantile of the residuals
  - $\bar{z}_\alpha$: the mean of the residuals $\hat{z}_t \le q_\alpha(\hat{z})$

Worked Forecast: S&P 500 on the Next Trading Day

- AR(1)-GARCH(1,1) on 2000–2026: $\hat{\phi} = -0.053$, $\hat{\alpha}_1 = 0.1204$, $\hat{\beta}_1 = 0.8604$; persistence $\hat{\alpha}_1 + \hat{\beta}_1 = 0.981$
  - Forecasts for the first trading day after 18 September 2026: $\hat{\mu}_{t+1} = 0.056\%$, $\hat{\sigma}_{t+1} = 0.725\%$ (sample standard deviation 1.21%: a calm market)
- Residual quantile $\hat{\xi}_{1\%} = -q_{1\%}(\hat{z}) = 2.79$, above $-z_{1\%} = 2.326$: the filtered shocks are still heavy-tailed
  - $\mathrm{VaR}_{1\%,t+1} = -0.056 + 0.725 \times 2.791 = 1.97\%$; $\mathrm{ES}_{1\%,t+1} = 2.55\%$
  - ES step: minus the mean of the residuals at or below $q_{1\%}(\hat{z})$ is 3.594 (Normal: 2.665), so $\mathrm{ES}_{1\%,t+1} = -0.056 + 0.725 \times 3.594 \approx 2.55\%$
- Unconditional historical VaR 1%: 3.45%; conditional numbers move with the market, the unconditional one does not
- GARCH with Normal shocks gives a VaR 1% of 1.63% for the same day (all five assets: Appendix)

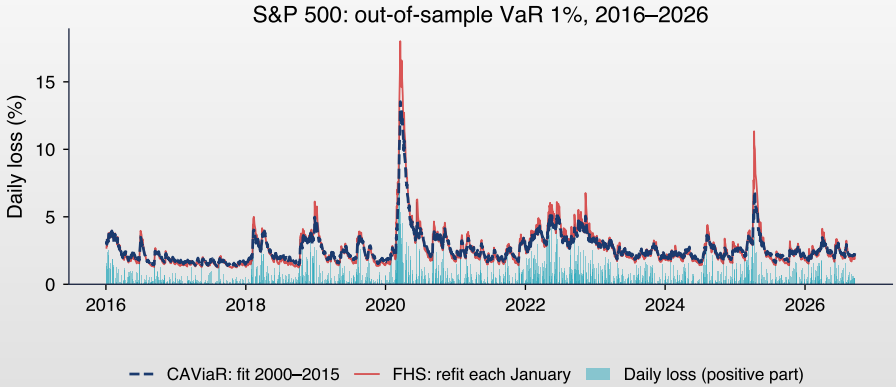
How Uncertain Is Tomorrow's VaR?

- Two estimated inputs: the GARCH parameters θ (through $\hat{\sigma}_{t+1}$) and the residual quantile $\hat{\xi}_\alpha$ Francq & Zakoïan (2015)
  - Residual bootstrap Christoffersen & Gonçalves (2005): simulate AR(1)-GARCH(1,1) paths with resampled $\hat{z}$, re-estimate on each path, forecast from the **original** history (algorithm in the Appendix)
- S&P 500, 999 paths: FHS $\mathrm{VaR}_{1\%} = 1.97\%$, 90% interval $[1.90, 2.09]\%$
- The two sources: $\hat{\sigma}_{t+1} = 0.725$ $[0.708, 0.743]$; $\hat{\xi}_{1\%} = 2.79$ $[2.68, 2.88]$
  - Only the quantile varies: $[1.89, 2.03]$; only σ_{t+1} varies: $[1.92, 2.02]$: the residual quantile (about 67 tail residuals) is the larger source
- GARCH with Normal shocks (1.63%) lies far below the interval: the choice of tail model matters more than the estimation error of a given model

Modelling the Quantile Directly: CAViaR

- CAViaR (Conditional Autoregressive Value at Risk) Engle & Manganelli (2004): the quantile follows its own recursion; SAV specification
  - $\mathrm{VaR}_t = \beta_0 + \beta_1 \mathrm{VaR}_{t-1} + \beta_2 |r_{t-1}|, \quad q_t = -\mathrm{VaR}_t$
  - SAV (symmetric absolute value): today's VaR = a constant β_0 + a share $\beta_1 \in (0,1)$ of yesterday's VaR (persistence) + $\beta_2 > 0$ times yesterday's absolute return
  - A large move of either sign raises VaR; q_t: the return quantile
- **Quantile regression** Koenker & Bassett (1978): minimise the pinball loss, with no assumption on the shock distribution
  - $\hat{\beta} = \arg\min_\beta \sum_t (\alpha - 1\{r_t < q_t(\beta)\})(r_t - q_t(\beta))$
  - $\arg\min_\beta$: the value of $\beta = (\beta_0, \beta_1, \beta_2)$ that minimises the sum; $q_t(\beta)$: the quantile path implied by β; elicitability turned into an estimator (optimiser settings in the Appendix)
- S&P 500, $\alpha = 1\%$, estimated on 2000–2015 (4 024 days): $\hat{\beta}_0 = 0.090$, $\hat{\beta}_1 = 0.897$, $\hat{\beta}_2 = 0.246$
  - Half-life of a shock to VaR: $\ln 0.5 / \ln \hat{\beta}_1 = 6.4$ days
- CAViaR reacts linearly to $|r_{t-1}|$, GARCH to r_{t-1}^2

CAViaR vs Filtered Historical Simulation, Out of Sample (1/2)



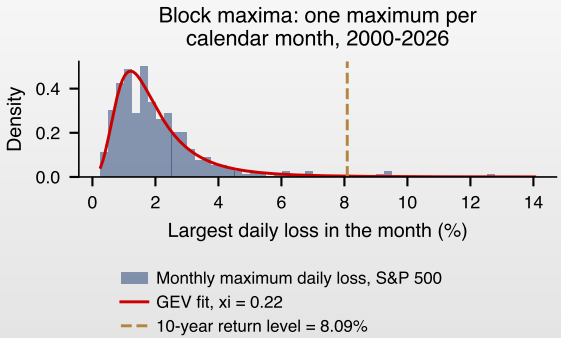
MFM_ch7_caviar

CAViaR vs Filtered Historical Simulation, Out of Sample (2/2)

- S&P 500, 2016–2026 (2 693 days): CAViaR with parameters fixed on 2000–2015; FHS re-estimated each January (a different refitting design)
- Exceedances of VaR 1%: CAViaR 1.30% (35), FHS 1.30% (35), historical 500 days 1.41% (38); mean pinball loss (lower is better): 0.039, 0.037, 0.052
- COVID-19, 20 February – 30 April 2020 (50 days): exceedances 6 / 3 / 9, peak VaR 13.5% / 18.0% / 4.5%: after a crash FHS jumps higher
- Counts alone do not establish that one model is better: significance is a Chapter 8 question

MFM_ch7_caviar

Monthly Maxima of S&P 500 Losses



- 321 calendar months, 2000–2026; GEV by maximum likelihood
- $\hat{\xi} = 0.22$ (Fréchet), $\hat{\mu} = 1.37$, $\hat{\sigma} = 0.79$: a power-law tail
- Return levels: 10 years 8.09%, 30 years 10.95%; largest monthly maximum 12.77%
- Monthly maxima do not give a daily VaR directly: the conversion assumes independent days, which clustering violates (Appendix)
- One value per month wastes data: the next method uses all large losses

MFM_ch7_gev

Block Maxima and the GEV Distribution (1/2)

- Split the sample into blocks of n days; keep each block maximum $M_n = \max(L_1, \ldots, L_n)$
  - Fisher–Tippett–Gnedenko theorem Fisher & Tippett (1928), Gnedenko (1943): if $(M_n - b_n)/a_n$ has a non-degenerate limit for i.i.d. losses, the limit is the generalised extreme value (GEV) distribution
  - $a_n > 0$, b_n: normalising constants that rescale and recentre the maximum
- The GEV distribution function: the probability that a block maximum is at most x
  - $H_\xi(x) = \exp\left\{-\left(1+\xi\,\frac{x-\mu}{\sigma}\right)^{-1/\xi}\right\}$, $1+\xi(x-\mu)/\sigma > 0$ ($\xi = 0$: $\exp\{-e^{-(x-\mu)/\sigma}\}$)
  - μ: location (where the maxima sit); $\sigma > 0$: scale (their spread); ξ: shape (the tail type, on the slide GEV (2/2); reused symbols: Appendix)

Block Maxima and the GEV Distribution (2/2)

- The **shape** ξ decides the tail
  - $\xi > 0$: Fréchet, power-law tail with tail index $1/\xi$ (financial returns)
  - $\xi = 0$: Gumbel domain, exponential-type tails (e.g. Normal, lognormal); $\xi < 0$: Weibull, bounded tail
- **Return level** for k blocks: the level exceeded on average once every k blocks, $H_\xi^{-1}(1 - 1/k)$, with H_ξ^{-1} the quantile function of the GEV
- In finance we model **only the tail** Embrechts, Klüppelberg & Mikosch (1997), Longin (2000)

Peaks over Threshold and the GPD (1/2)

- Fix a high threshold u; model the **excesses** $Y = L - u$ given $L > u$
  - Pickands–Balkema–de Haan theorem Pickands (1975), Balkema & de Haan (1974): for a wide class of distributions, as u approaches the right end-point of the distribution the excesses follow a generalised Pareto distribution
- The GPD distribution function: the probability that an excess is at most y
  - $G_{\xi,\beta}(y) = 1 - (1 + \xi y/\beta)^{-1/\xi}$ $(\xi \neq 0)$, $1 - e^{-y/\beta}$ $(\xi = 0)$
  - $y \geq 0$, and $y < -\beta/\xi$ if $\xi < 0$; scale $\beta > 0$: the typical size of an excess; shape ξ: the same as in the GEV, $\xi > 0$ = heavy tail

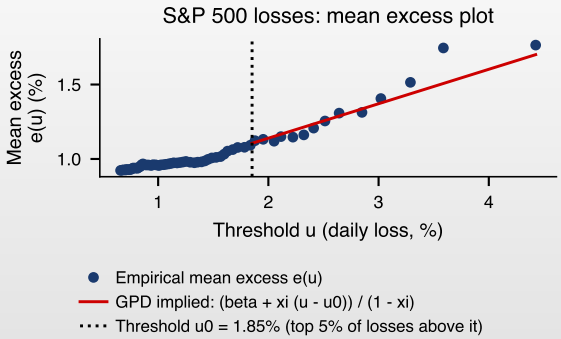
Peaks over Threshold and the GPD (2/2)

- Tail estimator with N_u exceedances among n observations McNeil, Frey & Embrechts (2015)
  - $\widehat{\mathrm{VaR}}_\alpha = u + \frac{\hat\beta}{\hat\xi}\left[\left(\frac{n}{N_u}\,\alpha\right)^{-\hat\xi} - 1\right], \qquad \widehat{\mathrm{ES}}_\alpha = \frac{\widehat{\mathrm{VaR}}_\alpha}{1-\hat\xi} + \frac{\hat\beta - \hat\xi u}{1-\hat\xi}$ ($\alpha < N_u/n$, $\hat\xi \neq 0$; ES needs $\hat\xi < 1$)
  - In words: VaR is the threshold u plus the GPD quantile of the excess that leaves probability α in the tail; N_u/n is the share of days above u
  - ES adds to VaR the mean excess beyond VaR, which grows with $\hat\xi$; $\hat\xi \geq 1$ means an infinite mean
- Bias–variance trade-off: a low u violates the limit theorem, a high u leaves few exceedances

Choosing the Threshold: the Mean Excess Function

- **Mean excess function**: $e(u) = \mathbb{E}[L - u \mid L > u]$; empirically, the average excess above u, plotted against u
- For a GPD tail $e(u)$ is **linear** in u
  - $e(u) = \dfrac{\beta + \xi(u - u_0)}{1 - \xi}$ for $u \geq u_0$, finite only if $\xi < 1$; slope $\xi/(1-\xi)$: upward for heavy tails, flat for exponential tails
  - u_0: the lowest threshold above which the GPD holds; β: the GPD scale at u_0
- Rule of thumb: choose u where the plot becomes roughly linear; check the stability of $\hat{\xi}$ across thresholds
- The Hill estimator of Chapter 1 Hill (1975) estimates the same tail index $1/\xi$

Mean Excess Plot: S&P 500



- Thresholds with 20% down to 0.5% of daily losses above them; red line: GPD fitted to the largest 5%
- The mean excess rises with u: a heavy (Fréchet-type) tail, not the flat line of an exponential tail
- Selected $u_0 = 1.85\%$ leaves $N_u = 336$ exceedances; empirical mean excess near u_0: 1.09% (GPD-implied $\hat{\beta}/(1-\hat{\xi}) = 1.11\%$; the GPD is fitted by maximum likelihood, not by matching this mean); GPD slope $\hat{\xi}/(1-\hat{\xi}) = 0.23$
- Beyond VaR 1% the points scatter: each averages very few losses

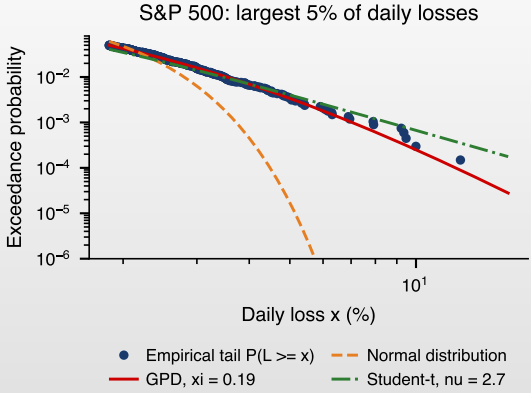
MFM_ch7_mean_excess

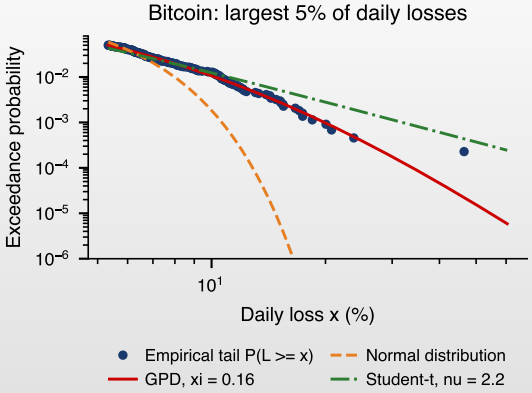
The GPD Tail Estimator on the S&P 500, Step by Step

- Inputs: $n = 6\,717$ daily losses, threshold $u = 1.852\%$, $N_u = 336$ exceedances; maximum likelihood $\hat{\xi} = 0.188$, $\hat{\beta} = 0.898$
- Step 1: $\dfrac{n}{N_u}\,\alpha = \dfrac{67.17}{336} = 0.200$, the share of the exceedances that lie beyond VaR 1%; $0.200^{-\hat{\xi}} = 1.353$
- Step 2: $\widehat{\mathrm{VaR}}_{1\%} = u + \dfrac{\hat{\beta}}{\hat{\xi}}\,(1.353 - 1) = 1.852 + 4.779 \times 0.353 \approx 3.54\%$
- Step 3: $\widehat{\mathrm{ES}}_{1\%} = \dfrac{3.540}{0.812} + \dfrac{0.898 - 0.188 \times 1.852}{0.812} = \dfrac{3.540 + 0.550}{0.812} \approx 5.04\%$
- Historical values for comparison: 3.45% and 5.01% (all five series: Appendix)

MFM_ch7_gpd_tail

GPD Tails: S&P 500 and Bitcoin (1/2)





MFM_ch7_gpd_tail

GPD Tails: S&P 500 and Bitcoin (2/2)

- Log–log scale: probability of a daily loss above x, largest 5% of losses; the Normal tail falls far below the data, the Student-t tail is close to the GPD
- S&P 500: $\hat{\xi} = 0.19$ (SE 0.065), tail index 5.3, $\mathrm{VaR}_{1\%} = 3.54\%$; Bitcoin: $\hat{\xi} = 0.16$ (SE 0.078), $\mathrm{VaR}_{1\%} = 10.23\%$ (all five series: Appendix)
- $\mathrm{SE}(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$ assumes independent exceedances; clustering makes it too small (next slide)

MFM_ch7_gpd_tail

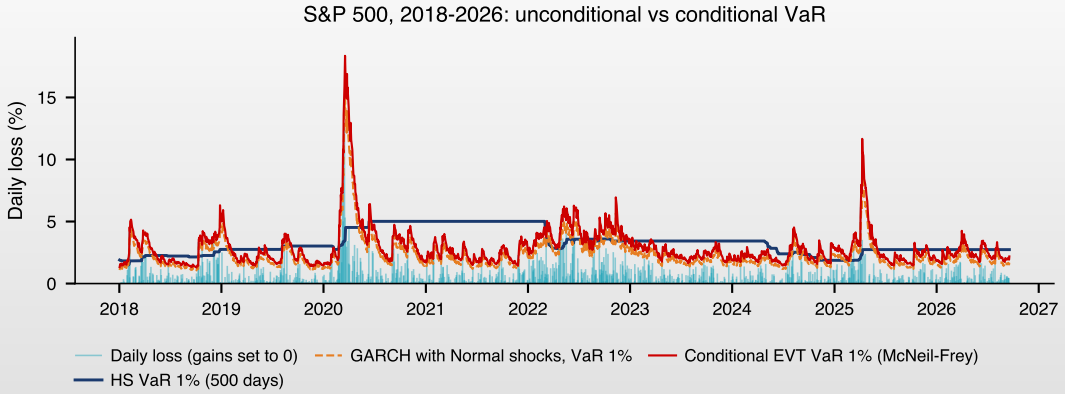
Clustered Extremes: the Extremal Index

- Stationary losses with clustering: if $nP(L > u_n) \to \tau$, then $P(M_n \leq u_n) \to e^{-\theta\tau}$, $\theta \in (0, 1]$ Embrechts, Klüppelberg & Mikosch (1997)
  - u_n: a threshold rising with n; τ: the expected number of exceedances of u_n in n days; M_n: the maximum of the n losses
  - θ = extremal index, $1/\theta$ = mean cluster size; i.i.d. data: $\theta = 1$; n dependent days act like $n\theta$ independent ones
- S&P 500, GPD threshold $u = 1.85\%$ ($N = 336$ exceedances): intervals estimator Ferro & Segers (2003) gives $\hat{\theta} = 0.19$, i.e. 62 clusters of 5.4 exceedances on average
  - GPD on cluster maxima: $\hat{\xi} = 0.152$ (SE 0.146) vs 0.188 (SE 0.065) on all exceedances: still positive, but its 95% interval now includes 0
- Standardised GARCH residuals: $\hat{\theta} = 0.99$: filtering removes the clustering of extremes (estimator and declustering: Appendix)
- This is why conditional EVT fits the GPD after the filter

Conditional EVT (McNeil and Frey)

- McNeil & Frey (2000): the FHS pipeline, with one step changed
- Filter: AR(1)-GARCH(1,1) by quasi-maximum likelihood, residuals $\hat{z}_t$, forecasts $\hat{\mu}_{t+1}$, $\hat{\sigma}_{t+1}$ (as in FHS)
- **Tail step**: instead of the empirical quantile of $-\hat{z}$, fit a GPD to its largest 10% and take its VaR $\hat{z}_\alpha^{GPD}$ and ES $\hat{e}_\alpha^{GPD}$
  - $\mathrm{VaR}_{\alpha,t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{z}_\alpha^{GPD}, \qquad \mathrm{ES}_{\alpha,t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{e}_\alpha^{GPD}$
  - $\hat{z}_\alpha^{GPD}$ takes the place of $\hat{\xi}_\alpha$ in the FHS formula: the same residual-loss quantile, now read from the fitted GPD tail of $-\hat{z}$
- S&P 500, first trading day after 18 September 2026: conditional EVT gives a VaR 1% of 2.03%, inside the FHS bootstrap interval $[1.90, 2.09]$%
- A smooth parametric tail beyond the data: matters most far in the tail (VaR 0.5%, VaR 0.1%), where only a handful of residuals exist

Conditional EVT on the S&P 500 (1/2)



MFM_ch7_cond_evt

Conditional EVT on the S&P 500 (2/2)

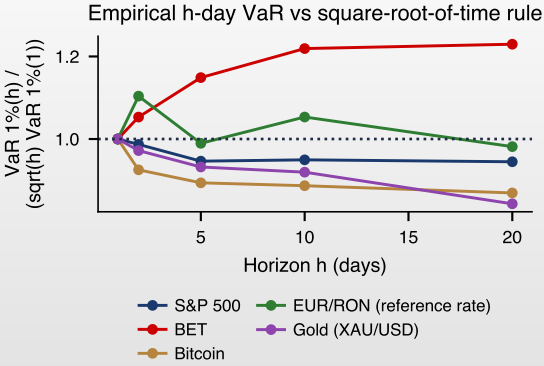
- One-day-ahead $\mathrm{VaR}_{1\%}$, 2018–2026, parameters re-estimated each January
- Exceedance rate (2 190 days, target 1%): conditional EVT 1.23%, GARCH with Normal shocks 2.42%, historical simulation over 500 days 1.60%
- GARCH with Normal shocks follows the rises and falls of risk correctly, but its VaR is too low (2.42% exceedances against 1%)
- Conditional EVT keeps the dynamics and corrects the level: the tail of the shocks matters, not only their scale

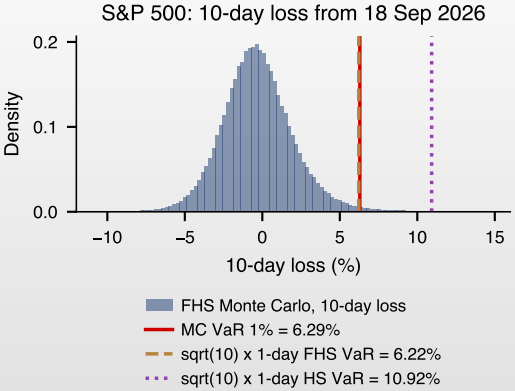
MFM_ch7_cond_evt

The Square-Root-of-Time Rule: Assumptions and Failures

- i.i.d. $N(0, \sigma^2)$ daily losses: $\mathrm{VaR}_\alpha^{(h)} = \sqrt{h}\,\mathrm{VaR}_\alpha^{(1)}$, $\mathrm{ES}_\alpha^{(h)} = \sqrt{h}\,\mathrm{ES}_\alpha^{(1)}$; Basel 1996 allowed $\sqrt{10} = 3.162$ Basel Committee (1996)
  - $\mathrm{VaR}_\alpha^{(h)}$, $\mathrm{ES}_\alpha^{(h)}$: the measures for the loss summed over h days; the standard deviation of a sum of h i.i.d. terms grows like $\sqrt{h}$
  - Needs zero mean, no autocorrelation, constant volatility and Normal shocks; each fails in the data (Chapters 1, 2, 5)
- **Autocorrelation** (variance ratio, Chapter 2)
  - $\mathrm{Var}(r_{t+1} + \cdots + r_{t+h}) = h\sigma^2\,\mathrm{VR}(h)$, with $\mathrm{VR}(h)$ the variance of the h-day sum divided by h times the daily variance
  - $\mathrm{VR} = 1$ without autocorrelation; $\mathrm{VR} > 1$ with positive autocorrelation, and then $\sqrt{h}$ **underestimates** the h-day risk
- **Volatility clustering**: volatility reverts to its mean over h days
  - Calm today $\Rightarrow \sqrt{h}$ too low; turbulent today $\Rightarrow$ too high
- **Heavy tails**: for i.i.d. shocks with tail index $a > 2$ the far-tail quantile of a sum grows like $h^{1/a} < \sqrt{h}$ Daníelsson (2011)
  - a: tail index, $P(L > x)$ decays like x^{-a}; smaller a = heavier tail
  - With jumps the rule can badly misstate multi-day risk Daníelsson & Zigrand (2006)

Square-Root-of-Time in the Data (1/2)





MFM_ch7_horizon

Square-Root-of-Time in the Data (2/2)

- Left: empirical $\mathrm{VaR}_{\mathbf{1\%}}$ of overlapping h-day losses divided by $\sqrt{h}$ times the one-day value (overlapping sums: far fewer independent observations); right: filtered Monte Carlo (MC in the legend), 200 000 paths
- Ratio at $h = 10$: BET 1.22 (lag-1 autocorrelation 0.109) and EUR/RON 1.05: $\sqrt{h}$ too low; S&P 500 0.95, gold 0.92, Bitcoin 0.89: conservative
- S&P 500 10-day $\mathrm{VaR}_{\mathbf{1\%}}$: filtered Monte Carlo 6.29%, $\sqrt{10}\times$ filtered one-day 6.22%, $\sqrt{10}\times$ historical one-day 10.92%: the big gap is conditional vs unconditional

MFM_ch7_horizon

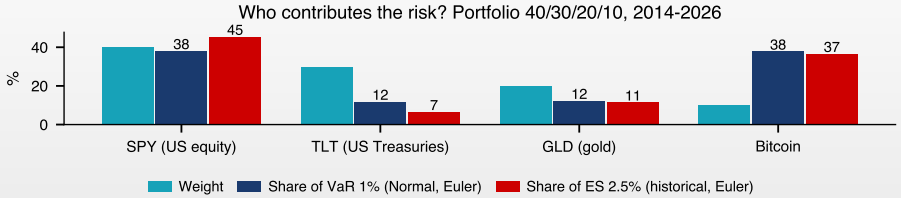
Filtered Monte Carlo over h Days

- Simulate many future paths from the fitted model and read VaR and ES from the simulated losses: handles autocorrelation, volatility clustering and heavy tails at once, as well as non-linear pay-offs (options)
- For each path $j = 1, \ldots, M$ (M: the number of simulated paths)
  - draw $z_{t+1}^{(j)}, \ldots, z_{t+h}^{(j)}$ with replacement from the standardised residuals $\hat{z}$
  - update $\sigma^{(j)}$ and $r^{(j)}$ day by day with the AR(1)-GARCH(1,1) recursion, starting from today's state
  - aggregate: $L^{(j)} = -\sum_{i=1}^{h} r_{t+i}^{(j)}$, the h-day loss of path j (i: day within the path)
- $\widehat{\mathrm{VaR}}_\alpha$, $\widehat{\mathrm{ES}}_\alpha$ = historical estimates on the M simulated losses; simulation error falls as $1/\sqrt{M}$

Quick Check: From One Day to Ten Days

- A desk reports a one-day $\mathrm{VaR}_{1\%} = 2\%$ from i.i.d. Normal losses with mean 0
  - (a) What is the 10-day $\mathrm{VaR}_{1\%}$ under the square-root-of-time rule?
  - (b) Returns turn out to be positively autocorrelated with $\mathrm{VR}(10) = 1.21$; returns stay Normal. What is the 10-day $\mathrm{VaR}_{1\%}$ now?
  - (c) No autocorrelation, but today's GARCH volatility is well below its long-run level. Is $\sqrt{10}\times$ the conditional one-day VaR too low or too high, and why?
- Answers in the ▸ Appendix

Who Contributes the Risk?



- SPY (S&P 500 ETF; ETF: exchange-traded fund) 40%, TLT (long-term US Treasury ETF) 30%, GLD (gold ETF) 20%, Bitcoin 10%, rebalanced daily; simple returns on 3 017 common days, 2014–2026; teal: weights, blue: shares of Normal $\mathrm{VaR}_{1\%}$, red: shares of historical $\mathrm{ES}_{2.5\%}$
- $\mathrm{VaR}_{1\%} = 1.77\%$ vs 3.11% undiversified (-43%); $\mathrm{ES}_{2.5\%} = 2.15\%$ vs 3.87% (-44%)
- Bitcoin: 10% of the money, 37.8% of VaR and 36.6% of ES; TLT: 30% of the money, only 6.5% of ES (it lost 0.47% on the portfolio's worst days)

MFM_ch7_components

Portfolio VaR and Euler Allocation (1/2)

- Setting: portfolio loss $L_p = -w^\top R$
  - w: vector of weights; R: vector of simple returns (simple returns add across assets), with covariance matrix Σ; σ_i: standard deviation of asset i
- Normal returns, mean ignored: VaR is a multiple of the portfolio standard deviation
  - $\mathrm{VaR}_\alpha(L_p) = z_{1-\alpha}\sigma_p, \quad \sigma_p = \sqrt{w^\top \Sigma w}$; $z_{1-\alpha} = -z_\alpha > 0$
  - For long-only $w \geq 0$, the undiversified VaR $\sum_i z_{1-\alpha} w_i \sigma_i$ (the sum of stand-alone VaRs) is larger unless all correlations are 1

Portfolio VaR and Euler Allocation (2/2)

- **Marginal VaR**: the change in portfolio VaR per unit increase of weight w_i
  - $\partial \mathrm{VaR}_\alpha / \partial w_i = z_{1-\alpha}(\Sigma w)_i / \sigma_p$; $(\Sigma w)_i$: the covariance of asset i with the portfolio
- **Component VaR** $C_i = w_i \, \partial \mathrm{VaR}_\alpha / \partial w_i$: the part of VaR attributed to asset i
  - Euler's theorem for a function homogeneous of degree 1: $\sum_i w_i \, \partial \rho / \partial w_i = \rho$, so the components add up, $\sum_i C_i = \mathrm{VaR}_\alpha$ Tasche (2008)
- ES, no distributional assumption (continuous losses): the mean loss of asset i on the portfolio's tail days
  - $C_i^{ES} = \mathbb{E}[-w_i R_i \mid L_p \geq \mathrm{VaR}_\alpha(L_p)]$, $\quad \sum_i C_i^{ES} = \mathrm{ES}_\alpha(L_p)$
  - Historical estimate: the average over the tail days; a negative C_i means the asset **hedges** the portfolio

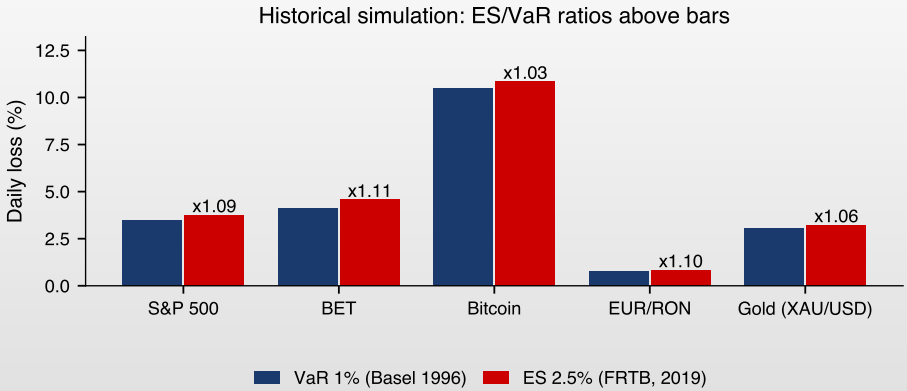
Minimum-ES Portfolio: a Linear Program

- With N historical scenarios R_s, the Rockafellar–Uryasev objective becomes a **linear program** Rockafellar & Uryasev (2000)
  - $\min_{w,v,u}\ v + \frac{1}{\alpha N}\sum_s u_s$ subject to $u_s \geq -w^\top R_s - v$, $u_s \geq 0$, $\sum_i w_i = 1$, $w \geq 0$
  - $s = 1, \ldots, N$: the scenarios; u_s: an auxiliary variable equal, at the optimum, to the excess $(-w^\top R_s - v)_+$ of scenario s over the threshold v; the optimal v is the portfolio VaR
  - Constraints: fully invested ($\sum_i w_i = 1$), no short sales ($w \geq 0$)
- Same four assets, 3 017 common days, $\alpha = 2.5\%$: weights 31% / 44% / 25% / 0%
  - $\mathrm{ES}_{2.5\%} = 1.67\%$, against 2.15% for the 40/30/20/10 portfolio
- Long-only minimum variance: 35% / 43% / 22% / 0%, $\mathrm{ES}_{2.5\%} = 1.67\%$: for these assets the two criteria almost agree; they part ways for options and credit
- Several portfolios can attain the same minimum: report the solution together with its tail scenarios

Liquidity-Adjusted VaR

- VaR assumes positions can be sold at the mid price; in reality we pay half the bid–ask spread (bid: the best price buyers offer; ask: the best price sellers accept; mid $= (\text{bid} + \text{ask})/2$)
  - **Relative spread** $S = (\text{ask} - \text{bid})/\text{mid}$; it widens exactly when markets fall
- Bangia, Diebold, Schuermann & Stroughair (2001): $\mathrm{LVaR}_\alpha = \mathrm{VaR}_\alpha + \frac{1}{2}(\mu_S + k\,\sigma_S)$
  - μ_S, σ_S: mean and standard deviation of the spread; k: a multiplier (here S is the spread, not skewness)
  - VaR plus the cost of crossing half a stressed spread, k standard deviations above its mean
- Banca Transilvania, a position of 1 million RON: the one-day historical VaR 1% over the last 496 days is 3.16%; with assumed $\mu_S = 0.5\%$, $\sigma_S = 0.3\%$, $k = 3$: $\mathrm{LVaR} = 3.86\%$ (+22%)
  - In money: VaR $\approx$ 31 581 RON, spread cost $\frac{1}{2}(0.5 + 3 \times 0.3)\% = 0.70\% \to 7\,000$ RON, LVaR $\approx$ 38 581 RON
- Large positions also move the price (endogenous liquidity): the regulator handles this with longer horizons

VaR 1% vs ES 2.5% on Five Assets (1/2)



MFM_ch7_basel

VaR 1% vs ES 2.5% on Five Assets (2/2)

- Historical simulation, daily losses, full samples; number above each pair: $\mathrm{ES}_{\mathbf{2.5\%}}/\mathrm{VaR}_{\mathbf{1\%}}$
- Normal benchmark: $2.338/2.326 \approx 1.005$, the same capital for thin tails; Student-t: 1.01 at $\nu = 30$, 1.07 at $\nu = 4$
- Data: S&P 500 1.09, BET 1.11, Bitcoin 1.03, EUR/RON 1.10, gold 1.06: the switch from VaR 1% to ES 2.5% raises the one-day measure by 3–11%
- ES 2.5% need not exceed VaR 1%: the inequality $\mathrm{ES}_\alpha \geq \mathrm{VaR}_\alpha$ is guaranteed only at the same level α
- Stressed calibration and liquidity horizons add more: Seminar 7, C2

MFM_ch7_basel

FRTB Terms: Desks, Horizons and the Stress Period

- **MAR**: the market-risk chapter of the consolidated Basel Framework; MAR12, MAR33.12 and so on are its paragraphs
- **Trading desk**: a defined group of traders and trading accounts with one business strategy; under FRTB internal models are approved desk by desk (MAR12)
- **Internal-models approach**: the bank computes market-risk capital with its own supervisor-approved ES model instead of the standardised formulas; **own-funds requirements**: the EU legal term for the capital it must hold
- **Liquidity horizon** (LH): the time needed to exit or hedge a position in stressed markets without moving its price; each risk factor gets 10, 20, 40, 60 or 120 days (MAR33.12)
- **Stressed period**: the 12-month window, searched back to at least 2007, that gives the current portfolio its largest ES; the ES is calibrated to it (MAR33.6–33.7)
- **Multiplier**: capital is the larger of the latest figure and $m_c\times$ the 60-day average, with $m_c = 1.5$ plus 0 to 0.5 after poor backtests (MAR33.41–33.42)
- **Teaching simplification** used in Seminar 7, C2: capital $\approx m_c \times \mathrm{ES}^{(\mathbf{10})}_{\mathbf{2.5\%}}\sqrt{\mathrm{LH}/10}$
  - Not the full MAR33 rule: the regulatory formula cascades over the LH buckets and adds further terms; the two agree on ES only when all risk factors share one LH

From Basel 1996 to FRTB (2019)

	Basel 1996 Basel Committee (1996)	FRTB 2019 Basel Committee (2019)
Risk measure	VaR 1%, computed daily	ES 2.5%, computed daily for every trading desk
Horizon	10 days; $\sqrt{10}$ scaling allowed	10-day base, scaled by liquidity horizons (large-cap equity 10, small-cap 20, credit spreads up to 120 days)
Data	at least one year of history	calibrated to the most severe 12-month stress period
Capital	max(yesterday's VaR, multiplier $\geq 3 \times$ 60-day average VaR); the multiplier rises by 0–1 after poor backtests (traffic-light zones, Chapter 8)	multiplier $m_c = 1.5$, increased after poor backtests
Backtesting	VaR 1% exceptions (Chapter 8)	one-day VaR 1% and VaR 2.5%: VaR tests, ES sets the capital

- Critique after 2008: VaR ignores the size of tail losses and one-year windows forget crises

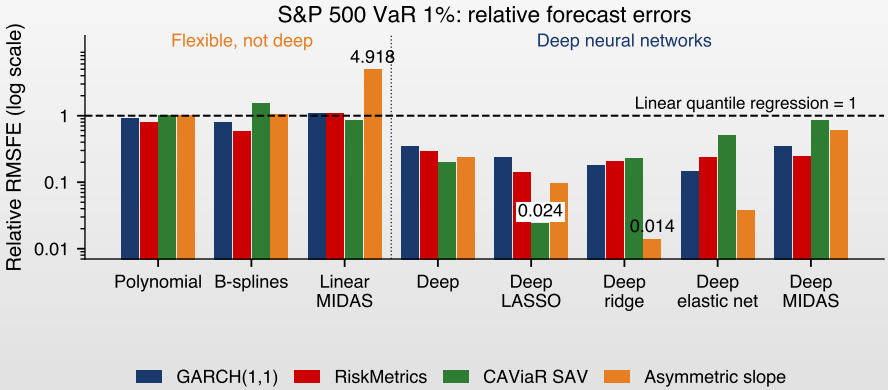
Case Study (1/4): Can a Deep Network Improve VaR Forecasts?

- **Question** Chronopoulos, Raftapostolos & Kapetanios (2024) (*Journal of Financial Econometrics*)
  - Does VaR forecasting improve if the conditional quantile may depend on its predictors in an **unknown non-linear** way, learned by a deep neural network?
- Data (paper, Section 3): S&P 500 daily log returns, September 1985 – August 2020; $\alpha = 1\%, 5\%, 10\%$
- Predictors x_t: GARCH or RiskMetrics volatility, CAViaR state variables
- Last 2 000 days out of sample, one fixed tuning window (details: Appendix)
- Benchmark: linear quantile regression Koenker & Bassett (1978) on the same predictors, i.e. a linear CAViaR-type specification

Case Study (2/4): the Conditional Quantile as a Neural Network

- Model (Eqs. 5–10): the conditional return quantile is a neural network of the predictors
  - $Q_{r_t}(\alpha \mid x_t) \approx G_\alpha(x_t, w)$
  - $Q_{r_t}(\alpha \mid x_t)$: the α-quantile of r_t given x_t; G_α: a feed-forward network with ReLU (rectified linear unit, $\max(\cdot, 0)$) hidden layers; w: its weights
- Estimation: minimise the pinball loss plus a penalty on w (none, LASSO = L_1 penalty, ridge = L_2 penalty, or elastic net = both); w^*: the fitted weights
- Target: the daily return quantile 10 days ahead (Eq. 23)
  - $\widehat{\mathrm{VaR}}_{t+10|\mathcal{F}_t} = -G_\alpha(x_t, w^*)$; $\mathcal{F}_t$: the information available at day t
- Evaluation: RMSFE (root mean squared forecast error) relative to the benchmark; below 1 = smaller error than linear quantile regression
  - Significance: Diebold–Mariano (DM) test of equal forecast accuracy

Case Study (3/4): Relative Forecast Error, $\alpha = 1\%$



MFM_ch7_crk_table6

Case Study (4/4): What the Results Show and Do Not Show

- Chronopoulos, Raftapostolos & Kapetanios (2024), Table 6: RMSFE relative to linear quantile regression (dashed line = 1), log scale; S&P 500, last 2 000 days to August 2020
- Flexible non-network models (polynomial, B-splines, linear MIDAS = mixed data sampling regression): 7 of 12 bars at or above 1; all 16 deep-network bars and the 4 deep MIDAS bars below 1 (0.014–0.850), all significant in the DM test except deep MIDAS for SAV
- Caution: RMSFE is minimised by the conditional **mean**, not the quantile; quantile accuracy needs the pinball loss
- One fixed tuning window: whether a 98% cut survives rolling re-estimation is open (Seminar 7, C3)

MFM_ch7_crk_table6

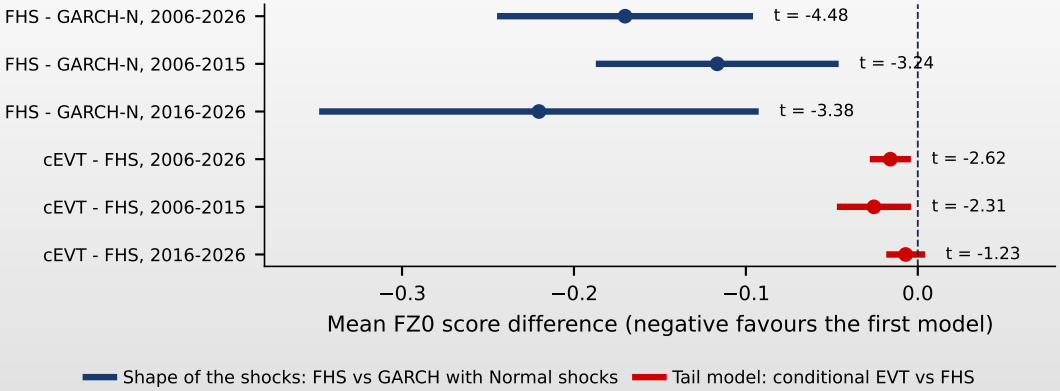
Can Daily Data Rank (VaR, ES) Models? (1/2): Score and Test

- Can daily data rank competing (VaR 1%, ES 1%) models? Nolde & Ziegel (2017), Patton, Ziegel & Chen (2019), Bayer & Dimitriadis (2022), Pele & Mazurencu-Marinescu-Pele (2026)
  - Obstacles: ES alone is not elicitable, and only about 1% of days are informative
- FZ0 score: a joint penalty for a (VaR, ES) forecast given the realised return, lower is better Patton, Ziegel & Chen (2019)
  - $S(v, e; Y) = -\frac{1}{\alpha e} 1\{Y \le v\}(v - Y) + \frac{v}{e} + \ln(-e) - 1$
  - Y: the realised return; $v_t = -\mathrm{VaR}_{\alpha,t}$ and $e_t = -\mathrm{ES}_{\alpha,t}$: the forecast quantile and tail mean, both negative; the first term penalises returns below v
- Test: $d_t = S_t^A - S_t^B$, the daily score difference of models A and B
  - H_0: $\mathbb{E}[d_t] = 0$ (equal accuracy) vs H_1: $\mathbb{E}[d_t] \neq 0$; Diebold–Mariano t-statistic with HAC errors, approximately $N(0, 1)$ under H_0; $t < -1.96$: A significantly better at 5%
  - We report the interval for $\mathbb{E}[d_t]$ separately on subperiods fixed before the analysis (2006–2015, 2016–2026)

Can Daily Data Rank (VaR, ES) Models? (2/2): What the Researcher Checks

- An AI assistant can draft the literature map, the code and a referee report; the researcher checks
  - Sign and level: VaR 1% $= -q_{1\%}$, positive for a loss (recompute one day of FZ0 step by step)
  - Look-ahead bias: each forecast uses only data before its day
  - Scoring: joint (VaR, ES) scores only, never the MSE (mean squared error) of ES alone Gneiting (2011)
  - Test power: count the tail days the comparison rests on
  - Every DOI (Digital Object Identifier) is resolved (reusable prompts: Appendix)

Mini-Case: Can We Rank ES Models? (1/2)



MFM_ch7_ai_discovery_case

Mini-Case: Can We Rank ES Models? (2/2)

- Claim tested: "conditional EVT is the best (VaR 1%, ES 1%) model for the S&P 500"; replicate first: FHS exceedance rate 1.54% (80 of 5 210 days, 2006–2026)
- Mean FZ0 (lower is better): conditional EVT 1.267, FHS 1.283, GARCH–Normal 1.453; bars: HAC 95% intervals for $\mathbb{E}[d_t]$; in the chart, cEVT = conditional EVT, GARCH-N = GARCH with Normal shocks
- All six Diebold–Mariano t-statistics (negative: the first model scores better; 5% critical value -1.96)
  - FHS vs GARCH–Normal: 2006–2026 $t = -4.48$; 2006–2015 $t = -3.24$; 2016–2026 $t = -3.38$
  - cEVT vs FHS: 2006–2026 $t = -2.62$; 2006–2015 $t = -2.31$; 2016–2026 $t = -1.23$
- Decision rule applied here: a gain counts as demonstrated only if it is significant at 5% in **both** pre-registered halves; the full-sample t is reported but is not decisive
- The shape of the shocks matters robustly: Normal shocks lose in both halves
- The EVT gain over FHS is ten times smaller, significant in 2006–2015 but not in 2016–2026 ($[-0.018, 0.004]$): by the rule above the gain is not demonstrated, and this does not prove the two models are equivalent

MFM_ch7_ai_discovery_case

Your Turn: A Team-Project Seed

- Question: does a dynamic semiparametric (VaR, ES) model Patton, Ziegel & Chen (2019) beat FHS and conditional EVT at ES 2.5%, out of sample?
  - Markets: the S&P 500, the BET and Bitcoin
  - Model to test: the one-factor GAS (generalised autoregressive score) model of Patton, Ziegel & Chen (2019) (their Section 2.3), estimated by minimising the FZ0 loss
  - Status: an optional extension for deeper study; the seminar team-project menu is Seminar 7, C2 and C3
- Replicate first: the S&P 500 FHS VaR 1% of 1.97% and the FZ0 ranking of the mini-case
- Deliverables
  - A one-page pre-registration: models, levels, scores, subperiods, power (tail days) and what would falsify the claim
  - Code on the daily market data of the course (starting point: Quantlet)
  - A table of all comparisons with adjusted p-values
  - A referee-style self-critique
- Log prompts and accepted outputs in `AI_USE.md`, every error caught (flipped sign, wrong level label, look-ahead bias, invented DOI) in `AI_ERRORS.md`

Which Method When?

Situation	Method	Fails when
Quick check, long stable history	Historical simulation, 500–1000 days	volatility changes; nothing beyond the worst loss
Daily risk after a volatility shock	FHS or conditional EVT, with a bootstrap interval	the GARCH model is misspecified
VaR 0.5% and deeper, stress tests	Peaks over threshold (GPD), conditional or not	threshold badly chosen; clustered exceedances
Multi-day horizon, options	Filtered Monte Carlo	model risk dominates
Risk budgeting across desks or assets	Euler contributions to ES	few tail days behind each contribution
Regulatory capital	ES 2.5%, 10 days, stressed period, VaR backtests	the stress window is short and noisy

- Every VaR or ES is an estimate: report it with its tail count and an interval

Review Questions and the Link to Chapter 8

Retrieval questions

- Why can VaR reward concentration?
- Why can ES not reward it?
- Why does the precision of ES depend on $n\alpha$ rather than on n?
- When is $\sqrt{10}$ scaling too low?
- When is it too high?

Chapter 8: backtesting

- How often did losses exceed VaR?
- Were the exceedances independent?
- Coverage tests and the Basel traffic-light zones
- Comparing (VaR, ES) forecasts with joint scores Fissler & Ziegel (2016)

Further reading: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), McNeil, Frey & Embrechts (2015)

Data Used in This Chapter

Series	Type	Period	Days
S&P 500	US equity index	2000–2026	6 717
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
EUR/RON	BNR reference rate	2005–2026	5 352
Gold	USD per troy ounce, weekdays	2000–2026	6 956
Portfolio	SPY, TLT, GLD (ETFs) and Bitcoin	2014–2026	3 017
Banca Transilvania	BVB share (liquidity example)	2024–2026	496

- Daily closing values up to 18 September 2026; daily log returns in %, $r_t = 100\,(\ln P_t - \ln P_{t-1})$
- Each series on its own calendar; the portfolio uses the days on which all four prices exist
- ETFs: prices adjusted for dividends; SPY tracks the S&P 500, TLT long US Treasury bonds, GLD gold

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Answers to the Quick Checks

- **Which position is riskier?**
  - A: $\mathrm{VaR}_{10\%} = q_{0.9}(L_A) = 9\%$; $\mathrm{ES}_{10\%}$ = mean of L_A on $[9, 10] = 9.5\%$
  - B: $P(L_B > 0) = 5\% \leq 10\%$, so $\mathrm{VaR}_{10\%} = 0$; $\mathrm{ES}_{10\%} = (20 \times 0.05 + 0 \times 0.05)/0.10 = 10\%$
  - VaR calls A riskier (9 vs 0), ES calls B riskier (10 vs 9.5): VaR never sees the 5% chance of losing 20
- **How many days make a precise VaR?**
  - (a) SE $\approx \sqrt{0.05 \times 0.95/1\,000}\,/\,0.1031 = 0.0069/0.1031 \approx 0.067$ pp; 95% half-width $1.96 \times 0.067 \approx 0.13$ pp
  - (b) SE falls like $1/\sqrt{n}$: $4 \times 1\,000 = 4\,000$ days, about 16 years of trading days
  - (c) $n\alpha = 10$ losses: a single extreme day moves the average a lot; precision is governed by $n\alpha$, not n
- **From one day to ten days**
  - (a) $2 \times \sqrt{10} = 6.32\%$; (b) $6.32 \times \sqrt{1.21} = 6.32 \times 1.1 \approx 6.96\%$
  - (c) Too low: volatility is expected to rise towards its long-run level during the ten days; S&P 500: filtered Monte Carlo 6.29% against $\sqrt{10}\times$ filtered one-day 6.22%

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Proof of the Rockafellar–Uryasev Representation

- **Proof**: $F(v) = v + \mathbb{E}[(L - v)_+]/\alpha$ is convex, with right derivative (slope just to the right of v) $1 - P(L > v)/\alpha$
  - It is ≥ 0 exactly when $P(L > v) \leq \alpha$, i.e. $v \geq \mathrm{VaR}_\alpha$
- **Corollary**: $(w, v) \mapsto v + \mathbb{E}[(-w^\top R - v)_+]/\alpha$ is jointly convex; minimising over v keeps convexity
  - ES is convex in w; with positive homogeneity this proves subadditivity
- Envelope argument: $\partial F/\partial v = 0$ at $v = \mathrm{VaR}_\alpha$, so a small error in VaR changes the minimum only to second order: the source of the ES standard error without a VaR term

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When Is VaR Subadditive After All?

- **Jointly elliptical** losses, $\alpha < 0.5$: VaR is subadditive (slide "Coherent Risk Measures (2/2)") McNeil, Frey & Embrechts (2015)
  - Normal or t marginals alone are not enough: the joint distribution must be elliptical
- **Heavy but not too heavy tails**: for multivariate regularly varying losses with tail index above 1, VaR is subadditive far enough in the tail (asymptotically, as $\alpha \to 0$) Daníelsson et al. (2013)
  - Estimated tail indices of equity and FX returns are in that range (between 4.8 and 6.5 in the Extreme Value Theory section), but the result is asymptotic, not a guarantee at a given α
- It fails for discrete jumps (defaults), extremely heavy tails and some option pay-offs
- VaR vs ES in practice: ES needs more data for the same precision Yamai & Yoshiba (2005)

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VaR Aggregation under Dependence Uncertainty

- Margins $F_1, \dots, F_d$ known, copula unknown (Chapter 6): $\mathrm{VaR}_\alpha(\sum_i L_i)$ lies in an interval Embrechts, Puccetti & Rüschendorf (2013)
  - F_i: distribution function of the loss L_i of asset i, d: number of assets; copula: the dependence structure that joins the margins
  - The comonotone value $\sum_i \mathrm{VaR}_\alpha(L_i)$ (all losses moving perfectly together) is **not** the upper end: VaR is not subadditive

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Rearrangement Algorithm and Model Risk

- **Rearrangement algorithm**: discretise the tail beyond each $\mathrm{VaR}_\alpha(L_i)$ in N points (one column per asset)
  - Order each column opposite to the sum of the others until the row sums are flat; worst VaR $\approx$ smallest row sum
- 40/30/20/10 portfolio of the Portfolio VaR section, daily $\mathrm{VaR}_{1\%}$, empirical margins, $N = 2\,000$
  - Historical dependence (empirical quantile of the portfolio loss) 1.95%; comonotone (sum of the empirical stand-alone VaRs) 3.63%; worst case 4.76% (1.31$\times$ comonotone)
  - These are empirical values; the Portfolio VaR section uses the Normal formula, hence its 1.77% and undiversified 3.11% for the same portfolio
- Capital based on VaR carries dependence model risk of the same order as the diversification benefit

VaR and ES by Method and Asset

	$\mathrm{VaR}_{1\%}$ (%)					$\mathrm{ES}_{2.5\%}$ (%)				
	Hist.	Normal	t	CF	EVT	Hist.	Normal	t	CF	EVT
S&P 500	3.45	2.80	3.46	6.08	3.54	3.76	2.81	3.95	6.65	3.78
BET	4.13	3.19	4.06	7.25	4.22	4.57	3.21	4.76	7.93	4.58
Bitcoin	10.51	7.99	11.06	18.89	10.23	10.84	8.03	13.37	20.69	10.84
EUR/RON	0.76	0.64	1.36	1.43	0.78	0.84	0.65	3.27	1.60	0.84
Gold	3.04	2.50	3.00	4.64	3.07	3.23	2.51	3.27	5.02	3.23

- Daily losses, full samples; CF = Cornish–Fisher, EVT = generalised Pareto tail fitted to the largest 5% of losses (the Extreme Value Theory section)
- Excess kurtosis K: S&P 500 10.7, BET 11.1, Bitcoin 12.0, EUR/RON 15.1, gold 7.7; Bitcoin needs about three times the S&P 500 buffer, EUR/RON about a fifth

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Asymptotic Theory for Historical VaR and ES

- **VaR**: for i.i.d. losses with density f continuous and positive at VaR_α,
  $\sqrt{n}\,(\widehat{\mathrm{VaR}}_\alpha - \mathrm{VaR}_\alpha) \to N\left(0, \dfrac{\alpha(1-\alpha)}{f(\mathrm{VaR}_\alpha)^2}\right)$
  - $\to$: convergence in distribution as $n \to \infty$; the variance of the limit is the asymptotic variance (avar), and SE $= \sqrt{\mathrm{avar}/n}$
- **ES** Chen (2008), with the influence function IF of the slide "Two Usable Standard Errors (2/2)
  - No term from estimating VaR: $\partial F/\partial v = 0$ at $v = \mathrm{VaR}_\alpha$; if $\mathbb{E}[(L - \mathrm{VaR}_\alpha)_+^2] < \infty$, the variance of the limit of $\sqrt{n}(\widehat{\mathrm{ES}}_\alpha - \mathrm{ES}_\alpha)$ is $\mathrm{avar} = \mathrm{Var}((L - \mathrm{VaR}_\alpha)_+)/\alpha^2$
  - Pareto-type tail with index $a > 2$, $\alpha \to 0$: $\mathrm{Var}((L - \mathrm{VaR}_\alpha)_+) \approx \frac{2\alpha \mathrm{VaR}_\alpha^2}{(a-1)(a-2)}$, so SE $\propto \mathrm{VaR}_\alpha/\sqrt{n\alpha}$ ($\propto$: proportional to) Pele & Mazurencu-Marinescu-Pele (2026)

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Precision on the S&P 500, Complete Table

	n	$\widehat{\mathrm{VaR}}_{1\%}$	SE i.i.d.	SE HAC	$\widehat{\mathrm{ES}}_{2.5\%}$	SE i.i.d.	SE HAC
2000–2026	6 717	3.45	0.13	0.19	3.76	0.15	0.26
Last 500 days	500	2.73	0.36	0.44	2.94	0.48	0.65

- Density $f(\mathrm{VaR})$ by a Gaussian kernel (Silverman bandwidth); HAC: Bartlett kernel, lag $\lfloor 4(n/100)^{2/9} \rfloor$ (10 and 5) Newey & West (1994)
- Full sample, $n = 6\,717$: SE of $\mathrm{VaR}_{1\%}$ under the Normal distribution with $\sigma = 1.2\%$ would be 0.055, under t_4 0.12; volatility clustering inflates the SE by 46–73% on the full sample and by 22–35% on the last 500 days

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GARCH Filters and Tomorrow's Volatility

	$\hat{\phi}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\alpha}_1 + \hat{\beta}_1$	$\hat{\sigma}$ (sample)	$\hat{\sigma}_{t+1}$	$-q_{1\%}(\hat{z})$
S&P 500	−0.053	0.1204	0.8604	0.981	1.21	0.73	2.79
BET	0.104	0.207	0.787	0.994	1.40	1.76	2.74
Bitcoin	−0.009	0.127	0.843	0.970	3.49	3.00	2.88
EUR/RON	0.089	0.148	0.852	1.000	0.28	0.08	2.80
Gold	−0.022	0.046	0.943	0.989	1.09	1.43	2.68

- $\hat{\sigma}_{t+1}$: forecast for the first trading day after 18 September 2026 (19 September for Bitcoin), in %
- Below 1, persistence implies mean reversion of variance forecasts. EUR/RON reaches the integrated GARCH boundary ($\hat{\alpha}_1 + \hat{\beta}_1 = 1$, within numerical precision): there is no finite unconditional variance, so variance forecasts do not revert to a finite long-run level
- $\hat{\sigma}$ (sample) is the empirical standard deviation of returns; it is not the model-implied long-run volatility
- $-q_{1\%}(\hat{z})$ between 2.68 and 2.88, above $-z_{1\%} = 2.326$: the filtered shocks are still heavy-tailed

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Conditional VaR and ES for the Next Trading Day

	$\mathrm{VaR}_{1\%}$ (%)				$\mathrm{ES}_{1\%}$ (%)			
	GARCH	FHS	c-EVT	Hist.	GARCH	FHS	c-EVT	Hist.
S&P 500	1.63	1.97	2.03	3.45	1.88	2.55	2.56	5.01
BET	3.80	4.53	4.63	4.13	4.39	6.16	6.20	6.28
Bitcoin	6.89	8.56	8.68	10.51	7.90	12.22	12.19	14.29
EUR/RON	0.18	0.22	0.21	0.76	0.21	0.28	0.28	1.12
Gold	3.32	3.82	3.86	3.04	3.80	4.94	4.93	4.21

- GARCH: GARCH with Normal shocks; c-EVT: conditional extreme value theory (the Extreme Value Theory section); Hist.: unconditional, full sample
- GARCH with Normal shocks is the lowest of the three conditional methods in every row: volatility alone does not create the tail

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Estimation Risk in Conditional VaR

- $\mathrm{VaR}_{\alpha,t+1} = -\mu_{t+1}(\theta) + \sigma_{t+1}(\theta)\,\xi_\alpha$, $\xi_\alpha = -q_\alpha(z)$: **both** θ (GARCH, by QML) and ξ_α (a residual quantile) are estimated
  - θ: the AR(1)-GARCH(1,1) parameters; QML: quasi-maximum likelihood; $\mu_{t+1}(\theta)$, $\sigma_{t+1}(\theta)$: the forecasts they imply
- Francq & Zakoïan (2015): estimate the **risk parameter** (θ, ξ_α) jointly; under strict stationarity, $\mathbb{E}[z_t^4] < \infty$ and a positive density of z at $q_\alpha(z)$, the conditional VaR is asymptotically Normal, with a variance from $\hat{\theta}$, from $\hat{\xi}_\alpha$ and from their covariance
- **Residual bootstrap** Christoffersen & Gonçalves (2005), for $b = 1, \ldots, B$ (B: number of bootstrap samples; a star marks a value computed on a bootstrap sample)
  - simulate an AR(1)-GARCH(1,1) path of the sample length with resampled $\hat{z}$ and re-estimate θ^* on it
  - σ^*_{t+1} from the **original** returns with θ^*; ξ^*_α from the path's own residuals; $\mathrm{VaR}^*_{t+1} = -\mu^*_{t+1} + \sigma^*_{t+1}\xi^*_\alpha$; percentile interval: the 5% and 95% quantiles of the B values VaR^*_{t+1}
- Heavy tails and skewness of z slow the convergence; Spierdijk (2016) compares interval methods for ARMA–GARCH VaR (ARMA: autoregressive moving average conditional mean) in exactly this case

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CAViaR Estimation Settings and Joint (VaR, ES) Models

- Estimation settings of Engle & Manganelli (2004): 10^4 random starts in $(0,1)^3$, the best 10 refined by simplex and quasi-Newton steps; the recursion starts at $\mathrm{VaR}_1 = -\hat{q}_\alpha$, minus the empirical α-quantile of the first 300 returns; these days stay in the estimation sample
- The pair (VaR, ES) can be **estimated** by minimising a strictly consistent joint score Fissler & Ziegel (2016); Chapter 8 uses the same scores to **evaluate** forecasts
  - Patton, Ziegel & Chen (2019): dynamic semiparametric (VaR, ES) models, e.g. with GAS (generalised autoregressive score) dynamics, estimated by minimising the FZ0 loss; consistency and asymptotic normality
  - Taylor (2019): CAViaR-type recursions for VaR and ES, estimated by an asymmetric Laplace likelihood
  - Dimitriadis & Bayer (2019): linear regression of the quantile and of ES on covariates (realised and implied volatility), with standard errors
- Common thread: the scoring function that ranks forecasts is also the objective function that estimates the model

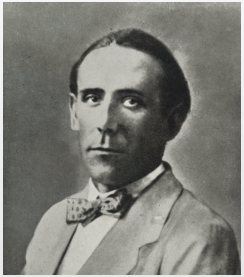
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One Symbol, Several Meanings: Extreme Value Notation

- ξ: the GEV/GPD shape; $\hat{\xi}_\alpha$ (with a level): the FHS residual quantile $-q_\alpha(\hat{z})$, which conditional EVT estimates from a GPD and writes $\hat{z}_\alpha^{GPD}$
- β: GPD scale; β_1: GARCH coefficient; $\beta_0, \beta_1, \beta_2$: CAViaR coefficients
- μ, σ: GEV location and scale in the block-maxima section; elsewhere the conditional mean and volatility
- u: the GPD threshold; in the ES definition, u is the integration variable; u_s: the auxiliary variable of the minimum-ES linear program
- α: tail probability; α_1: GARCH coefficient; $a = 1/\xi$: tail index

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Emil Gumbel and the Statistics of Extremes



Emil Julius Gumbel (1891–1966)

Photo: unknown author (c. 1931); public domain; Wikimedia Commons

- Fisher & Tippett (1928): the maximum of many i.i.d. variables, suitably rescaled, can have only three limit forms
- Gnedenko (1943) proved the result rigorously
- Gumbel (1958) wrote the first textbook and applied it to floods, droughts and life lengths
  - The Gumbel distribution is one of the three limit forms

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From Monthly Maxima to a Daily VaR

- Independent days: $P(M_n \leq x) \approx P(L \leq x)^n$, so $\mathrm{VaR}_\alpha \approx H^{-1}((1-\alpha)^n)$ with $n = 20.9$ trading days per month and H^{-1} the quantile function of the fitted GEV
  - $\mathrm{VaR}_{1\%} \approx H^{-1}(0.810) = 2.83\%$ vs historical 3.45%; $\mathrm{VaR}_{0.1\%} \approx 6.19\%$
- Volatility clustering breaks the independence assumption, and the approximation is too low here
- With the extremal index: $P(M_n \leq x) \approx F(x)^{n\theta}$, F: the distribution function of the daily loss
  - $\hat{\theta} = 0.35$ (threshold at VaR 1%): $\mathrm{VaR}_{1\%} \approx H^{-1}(0.929) = 4.14\%$ instead of 2.83%; with $\hat{\theta} = 0.19$: 5.09%
  - The correction moves the answer by more than the error it fixes: the block-maxima route to a daily VaR is fragile

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GPD Fits for Five Series

	u (%)	N_u	$\hat{\xi}$	SE($\hat{\xi}$)	$\hat{\beta}$	$1/\hat{\xi}$	$\mathrm{VaR}_{1\%}$	$\mathrm{ES}_{1\%}$
S&P 500	1.85	336	0.19	0.065	0.898	5.3	3.54	5.04
BET	1.93	334	0.21	0.066	1.196	4.8	4.22	6.34
Bitcoin	5.33	220	0.16	0.078	2.651	6.1	10.23	14.36
EUR/RON	0.38	268	0.18	0.072	0.217	5.5	0.78	1.14
Gold	1.68	348	0.15	0.062	0.759	6.5	3.07	4.22

- $\hat{\xi}$ between 0.15 and 0.21 for all five series, each more than two i.i.d. standard errors above 0: heavy tails everywhere (clustering makes these SEs too small)
- Tail index $1/\hat{\xi}$ between 4.8 and 6.5: the fitted upper tails have finite fourth moments, but every 95% interval for ξ reaches above 0.25, so finite fourth moments cannot be confirmed (compare Chapter 1)

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The Extremal Index: Estimator and Declustering

- **Intervals estimator** Ferro & Segers (2003) from the N exceedance times of u and the gaps T_i between them
  - $\hat{\theta} = \min\left\{1, \dfrac{2\left(\sum_i (T_i - 1)\right)^2}{(N-1)\sum_i (T_i - 1)(T_i - 2)}\right\}$ if $\max T_i > 2$ (else with T_i, T_i^2); it compares the squared mean gap with the mean squared gap: short gaps inside clusters push $\hat{\theta}$ below 1
- **Declustering** Ferro & Segers (2003): target $C = \lfloor \hat{\theta} N \rfloor + 1$ clusters, separated by the $C - 1$ largest gaps; fit the GPD to the cluster maxima
  - Ties: if the $(C-1)$-th and C-th largest gaps are equal, C is reduced until they differ, so the realised number of clusters can be below the target
  - The i.i.d. SE of $\hat{\xi}$ was less than half the SE from the cluster maxima
- S&P 500: $\hat{\theta} N = 0.18738 \times 336 = 62.96$, so target $C = \lfloor 62.96 \rfloor + 1 = 63$ (use the unrounded $\hat{\theta}$; the 0.19 of the main slide is rounded), 62 clusters after the tie rule; at the historical VaR 1% ($u = 3.45\%$, $N = 68$): $\hat{\theta} = 0.35$

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The Portfolio: Volatilities, Correlations and Contributions

	Weight	σ_i (%)	corr. SPY	corr. TLT	corr. GLD
SPY	40%	1.10	1.00		
TLT	30%	0.93	−0.17	1.00	
GLD	20%	1.03	0.07	0.25	1.00
Bitcoin	10%	4.14	0.23	−0.02	0.09

	σ_i (%)	Marginal VaR	C_i	share	C_i^{ES}	share
SPY	1.10	1.68	0.67	38.1%	0.98	45.4%
TLT	0.93	0.70	0.21	11.8%	0.14	6.5%
GLD	1.03	1.08	0.22	12.3%	0.25	11.4%
Bitcoin	4.14	6.68	0.67	37.8%	0.79	36.6%
Total	$\sigma_p = 0.76$		1.77	100%	2.15	100%

- SPY carries more of the ES than of the Normal VaR (45.4% vs 38.1%): its losses concentrate on the portfolio's worst days more than covariances imply; calling this tail dependence needs a separate analysis (Chapter 6)

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The Basel Committee and the EU Timetable



The tower of the Bank for International Settlements (BIS) in Basel, seat of the Basel Committee on Banking Supervision

Photo: Michal Pleskowicz (2018); CC BY-SA 4.0; Wikimedia Commons

- The Basel Committee on Banking Supervision sets global capital standards; national supervisors (in the EU, via the Capital Requirements Regulation) turn them into law
- **1996**: market risk enters the capital rules; banks may use internal VaR models Basel Committee (1996)
- **2019**: the Fundamental Review of the Trading Book standard replaces VaR by ES in the internal-models approach Basel Committee (2019); each jurisdiction sets its own application date
- EU: Delegated Regulation (EU) 2025/1496 postponed the FRTB own-funds requirements to 1 January 2027; until then banks apply the earlier VaR-based framework

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Case Study, Predictors, Sample and Tuning (1/2)

- **Data** (paper, Section 3.1): S&P 500 daily log returns $r_t = \ln(P_t/P_{t-\mathbf{1}})$, September 1985 – August 2020, $T = 9\,053$; $\alpha =$ 1%, 5% and 10%
  - Four predictor sets x_t (Eqs. 14–17, following Giacomini & Komunjer (2005)): GARCH(1,1) σ_t; RiskMetrics σ_t with decay factor $\lambda = 0.94$; CAViaR SAV $(\mathrm{VaR}_{t-\mathbf{1}}, |r_{t-\mathbf{1}}|)$; CAViaR asymmetric slope $(\mathrm{VaR}_{t-\mathbf{1}}, r^+_{t-\mathbf{1}}, r^-_{t-\mathbf{1}})$, $r^+ = \max(r, 0)$, $r^- = \max(-r, 0)$
  - The paper writes τ for our α; $\mathrm{VaR}_{\alpha,t} = -Q_{r_t}(\alpha \mid x_t)$, as in the VaR and Expected Shortfall section
- **Design** (paper, Section 3.2): the last 2 000 days are out of sample; the first 7 053 tune the model once (fixed forecast window)
  - Time-series CV (cross-validation, Eq. 12): gap $\delta = 5$ days around each left-out point, $\kappa = 20$ random test points; depth and width in $\{1, 5, 10\}$, learning rate and the penalty weight λ (not the RiskMetrics λ) five values in $[0.001, 0.01]$, batch size $\{10, 20\}$, dropout up to 20%, Adam optimiser (stochastic gradient descent with adaptive step sizes), early stopping (paper, Section 1.4)

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Case Study, Rival Models and Evaluation Tests (2/2)

- **Rivals**: linear quantile regression (the benchmark), quantile polynomial and B-splines, linear MIDAS (Eq. 18) and a deep MIDAS network
  - **Evaluation** (paper, Section 3.3): RMSFE relative to the linear benchmark; DM test Diebold & Mariano (1995) on quantile losses with the correction of Harvey, Leybourne & Newbold (1997); CQFE (conditional quantile forecast encompassing) test Giacomini & Komunjer (2005): does one forecast already contain the information of the other; smoothing $\eta = 0.005$
  - Motivation: Patton, Ziegel & Chen (2019) extended CAViaR to (VaR, ES) with parametric dynamics; this paper keeps VaR alone but drops linearity

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Case Study, Complete Results (Tables 6 and 7)

$\alpha = 1\%$	Polyn.	Splines	MIDAS	Deep	LASSO	Ridge	Elnet	Deep MIDAS
GARCH	0.909	0.791	1.083	0.346	0.236	0.177	0.146	0.346
RiskMetrics	0.780	0.570	1.066	0.289	0.141	0.207	0.239	0.246
SAV	1.000	1.511	0.838	0.198	0.024	0.225	0.498	0.850
Asym. slope	1.000	1.040	4.918	0.235	0.096	0.014	0.037	0.601

- RMSFE relative to linear quantile regression, $\alpha = 1\%$ (Table 6; LASSO, Ridge, Elnet = penalised deep networks): all 16 deep entries are below 1 and significant at 1% in the DM test
  - Over all levels, the deep networks cut the error by 50%–98%, the deep MIDAS by 11%–84%; the best penalty changes with the predictor set (elastic net for GARCH, LASSO for RiskMetrics and SAV, ridge for the asymmetric slope)
- In most cases the linear benchmark has smaller errors than the polynomial and spline regressions (ratios up to 1.716 at other α levels in Table 6); linear MIDAS reaches 4.918; the results do not isolate the contributions of architecture, penalisation and CV
- CQFE at $\alpha = 1\%$ (Table 7): the non-linear block encompasses its rivals 757 times (42 losses, 751 inconclusive); the linear block 173 times (15, 170); many inconclusive pairs mean that combining forecasts could still add information

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Case Study, Replication and Limitations

- **Replicate** on the daily market data of the course: the S&P 500 series starts in January 1990, so 1985–1989 is missing (a data limit)
  - Keep the evaluation window of the paper (the last 2 000 returns up to 31 August 2020, from 19 September 2012): 5 726 tuning days instead of 7 053
  - Build the GARCH and SAV predictors, fit linear and deep quantile regressions at $\alpha = 1\%$ and recompute these two rows of Table 6 (Seminar 7, C3)
- **Limits stated by the authors**
  - One fixed estimation window: recursive or rolling re-tuning over 2 000 days is computationally challenging (paper, Section 3.2)
  - Networks overfit easily, hence penalties, dropout, CV and early stopping (paper, Section 1.4); the output is read only through SHAP (Shapley additive explanations) values and partial derivatives (paper, Section 4)
- **Open extensions** (Seminar 7, C3)
  - Does the gain survive rolling re-estimation on 2020–2026 and the BET?
  - Can a network trained on the FZ0 loss of Patton, Ziegel & Chen (2019) forecast (VaR, ES) jointly, as FRTB needs?

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The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**: *"List papers since 2015 on comparative backtesting of (VaR, ES) with strictly consistent scoring functions: DOI, scoring function, test, sample size and power findings."*
  - Resolve every DOI on Crossref; check the sign convention of the score in the paper
- **Hypotheses**: *"For FHS, conditional EVT and GARCH with Normal shocks, state which features of the tail each captures and what each predicts for the FZ0 differences in calm and in turbulent years."*
- **Data and code**: *"Write Python that computes the FZ0 loss for daily VaR 1% and ES 1% forecasts on the S&P 500 from the daily market data of the course, with the sign convention VaR $= -q_\alpha$, and a Diebold–Mariano test with HAC errors."*
  - Replicate first: the FHS exceedance rate 1.54% over 5 210 days (the Conditional VaR and ES section)

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**: *"Write a pre-registered plan: subperiods 2006–2015 and 2016–2026, levels 1% and 2.5%, the FZ0 and a second consistent score, HAC lags 5 and 20, a second index."*
  - Size check: simulate from a fitted GARCH-t and score two forecasts with equal expected FZ0; the test must reject in about 5% of simulations
- **Critique**: *"Act as a referee: is my ranking driven by 2008 and 2020, by the score's sign convention or by overlapping estimation windows? Give the test that would expose each."*
- **Report**: every model pair, level and period; the number of tail days behind each comparison
- **Multiple testing**: models $\times$ levels $\times$ periods $\times$ assets: fix the family in advance and adjust the p-values; an unresolvable DOI or a misattributed score: drop the claim and log it in `AI_ERRORS.md`
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

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Quantinar Courses and the Chapter Code

Advanced courses on quantinar.com

- Measuring Statistical Risk
- XFG Advanced Methods in Quantitative Finance

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_07

Acronyms Used in This Material (1/2)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIS**: Bank for International Settlements
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CF**: Cornish–Fisher (expansion)
- **COVID-19**: COronaVIrus Disease 2019
- **CQFE**: Conditional Quantile Forecast Encompassing (test)
- **CV**: Cross-Validation
- **DM**: Diebold–Mariano (test)
- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund
- **EU**: European Union
- **EVT**: Extreme Value Theory
- **FHS**: Filtered Historical Simulation
- **FRTB**: Fundamental Review of the Trading Book
- **FX**: Foreign eXchange

Acronyms Used in This Material (2/2)

- **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-N**: GARCH with Normal innovations
- **GAS**: Generalised Autoregressive Score (model)
- **GEV**: Generalised Extreme Value (distribution)
- **GPD**: Generalised Pareto Distribution
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HS**: Historical Simulation
- **IF**: Influence Function
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MC**: Monte Carlo (simulation)
- **MIDAS**: MIxed DAta Sampling (regression)
- **MSE**: Mean Squared Error
- **QML**: Quasi-Maximum Likelihood
- **RMSFE**: Root Mean Squared Forecast Error
- **SAV**: Symmetric Absolute Value (CAViaR specification)
- **SE**: Standard Error
- **SHAP**: SHapley Additive exPlanations
- **US**: United States
- **XAU**: gold, ISO 4217 code (one troy ounce)
- **XFG**: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

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